

# TCA9548A Low-Voltage 8-Channel I<sup>2</sup>C Switch with Reset

## 1 Features

- 1-to-8 Bidirectional translating switches
- I<sup>2</sup>C Bus and SMBus compatible
- Active-low reset input
- Three address pins, allowing up to eight TCA9548A devices on the I<sup>2</sup>C bus
- Channel selection through an I<sup>2</sup>C Bus, in any combination
- Power up with all switch channels deselected
- Low R<sub>ON</sub> switches
- Allows voltage-level translation between 1.8-V, 2.5-V, 3.3-V, and 5-V buses
- No glitch on power up
- Supports hot insertion
- Low standby current
- Operating power-supply voltage range of 1.65 V to 5.5 V
- 5-V Tolerant inputs
- 0- to 400-kHz Clock frequency
- Latch-up performance exceeds 100 mA Per JESD 78, class II
- ESD Protection exceeds JESD 22
  - ±2000-V Human-body model (A114-A)
  - 200-V Machine model (A115-A)
  - ±1000-V Charged-device model (C101)

## 2 Applications

- Servers
- Routers (telecom switching equipment)
- [Factory Automation](#)
- Products with I<sup>2</sup>C slave address conflicts (such as multiple, identical temperature sensors)

## 3 Description

The TCA9548A device has eight bidirectional translating switches that can be controlled through the I<sup>2</sup>C bus. The SCL/SDA upstream pair fans out to eight downstream pairs, or channels. Any individual SCn/SDn channel or combination of channels can be selected, determined by the contents of the programmable control register. These downstream channels can be used to resolve I<sup>2</sup>C slave address conflicts. For example, if eight identical digital temperature sensors are needed in the application, one sensor can be connected at each channel: 0-7.

The system master can reset the TCA9548A in the event of a time-out or other improper operation by asserting a low in the **RESET** input. Similarly, the power-on reset deselects all channels and initializes the I<sup>2</sup>C/SMBus state machine. Asserting **RESET** causes the same reset and initialization to occur without powering down the part. This allows recovery should one of the downstream I<sup>2</sup>C buses get stuck in a low state.

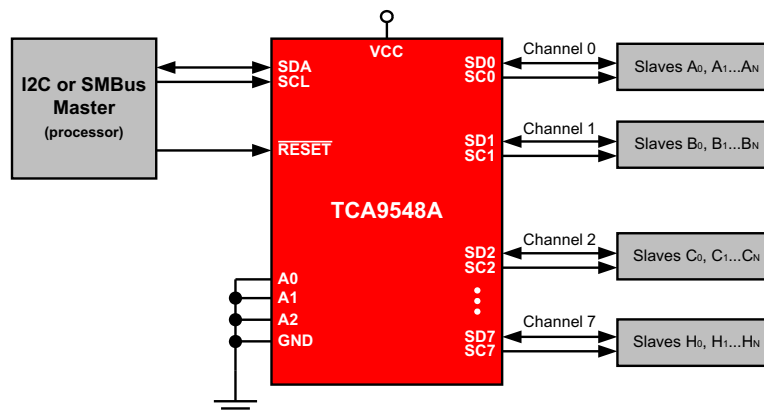
The pass gates of the switches are constructed so that the VCC pin can be used to limit the maximum high voltage, which is passed by the TCA9548A. Limiting the maximum high voltage allows the use of different bus voltages on each pair, so that 1.8-V, 2.5-V or 3.3-V parts can communicate with 5-V parts, without any additional protection. External pullup resistors pull the bus up to the desired voltage level for each channel. All I/O pins are 5-V tolerant.

### Device Information<sup>(1)</sup>

| PART NUMBER | PACKAGE    | BODY SIZE (NOM)   |
|-------------|------------|-------------------|
| TCA9548A    | TSSOP (24) | 7.80 mm × 4.40 mm |
|             | VQFN (24)  | 4.00 mm × 4.00 mm |

(1) For all available packages, see the orderable addendum at the end of the data sheet.

### Simplified Application Diagram



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## 4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision F (November 2016) to Revision G                                                                                            | Page |
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| • Changed the appearance of the PW package and the RGE package images .....                                                                      | 4    |
| • Changed T <sub>J</sub> from 90 C to 130 C in lower voltage V <sub>CC</sub> conditions .....                                                    | 5    |
| • Changed T <sub>A</sub> from 85 C to 125C for lower voltage V <sub>CC</sub> conditions .....                                                    | 5    |
| • Changed From: V <sub>CC</sub> = 2.3 V to 3.6 V To: V <sub>CC</sub> = 1.65 V to 5.5 V in the <i>Electrical Characteristics</i> conditions ..... | 6    |
| • Changed V <sub>O</sub> min from 0.9V to 0.6 V .....                                                                                            | 6    |
| • Added standby mode specifications for > 85 C T <sub>A</sub> .....                                                                              | 6    |
| • Changed R <sub>L</sub> = 1 kW To: R <sub>L</sub> = 1 KΩ in <a href="#">Figure 6</a> .....                                                      | 11   |

| Changes from Revision E (October 2015) to Revision F    | Page |
|---------------------------------------------------------|------|
| • Updated the <a href="#">Description</a> section ..... | 1    |
| • Added new orderable part number, TCA9548AMRGER .....  | 1    |

| Changes from Revision D (January 2015) to Revision E                   | Page |
|------------------------------------------------------------------------|------|
| • Updated Pin Functions table. ....                                    | 4    |
| • Added new I <sup>2</sup> C Sections and read/write description ..... | 16   |

| Changes from Revision C (November 2013) to Revision D                                                                                                                                                                                                                                                                                                                                                | Page |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| • Added <i>Pin Configuration and Functions</i> section, <i>ESD Ratings</i> table, <i>Feature Description</i> section, <i>Device Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section ..... | 1    |
| • Updated Typical Application schematic. ....                                                                                                                                                                                                                                                                                                                                                        | 21   |

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|--------------------------------------------------------------|-------------------|
| • Updated $V_{POR}$ and $I_{CC}$ standby specification. .... | <a href="#">6</a> |
|--------------------------------------------------------------|-------------------|
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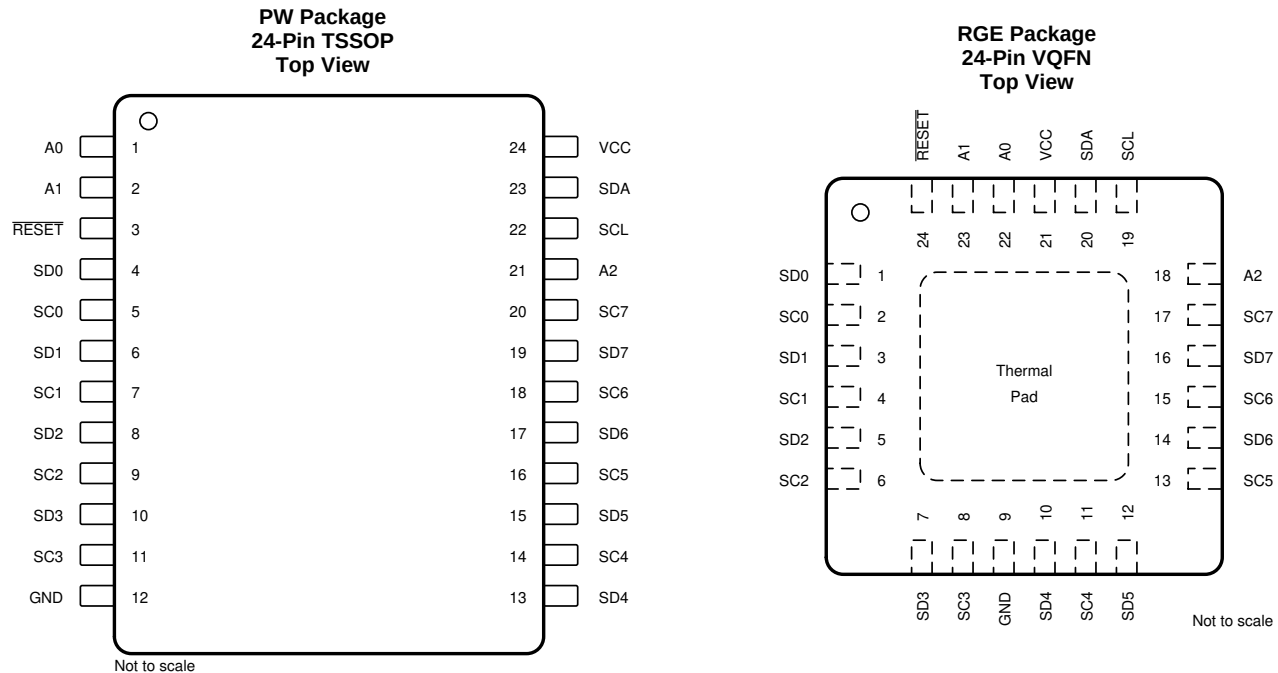
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| <b>Changes from Revision A (July 2012) to Revision B</b> | <b>Page</b> |
|----------------------------------------------------------|-------------|

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|--------------------------------------|-------------------|
| • Updated document formatting. ....  | <a href="#">1</a> |
| • Removed ordering information. .... | <a href="#">1</a> |
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## 5 Pin Configuration and Functions



### Pin Functions

| NAME  | PIN        |           | TYPE  | DESCRIPTION                                                                                                                    |
|-------|------------|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------|
|       | TSSOP (PW) | QFN (RGE) |       |                                                                                                                                |
| A0    | 1          | 22        | I     | Address input 0. Connect directly to V <sub>CC</sub> or ground                                                                 |
| A1    | 2          | 23        | I     | Address input 1. Connect directly to V <sub>CC</sub> or ground                                                                 |
| A2    | 21         | 18        | I     | Address input 2. Connect directly to V <sub>CC</sub> or ground                                                                 |
| GND   | 12         | 9         | —     | Ground                                                                                                                         |
| RESET | 3          | 24        | I     | Active-low reset input. Connect to V <sub>CC</sub> or V <sub>DPU0</sub> <sup>(1)</sup> through a pull-up resistor, if not used |
| SD0   | 4          | 1         | I/O   | Serial data 0. Connect to V <sub>DPU0</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC0   | 5          | 2         | I/O   | Serial clock 0. Connect to V <sub>DPU0</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD1   | 6          | 3         | I/O   | Serial data 1. Connect to V <sub>DPU1</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC1   | 7          | 4         | I/O   | Serial clock 1. Connect to V <sub>DPU1</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD2   | 8          | 5         | I/O   | Serial data 2. Connect to V <sub>DPU2</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC2   | 9          | 6         | I/O   | Serial clock 2. Connect to V <sub>DPU2</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD3   | 10         | 7         | I/O   | Serial data 3. Connect to V <sub>DPU3</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC3   | 11         | 8         | I/O   | Serial clock 3. Connect to V <sub>DPU3</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD4   | 13         | 10        | I/O   | Serial data 4. Connect to V <sub>DPU4</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC4   | 14         | 11        | I/O   | Serial clock 4. Connect to V <sub>DPU4</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD5   | 15         | 12        | I/O   | Serial data 5. Connect to V <sub>DPU5</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC5   | 16         | 13        | I/O   | Serial clock 5. Connect to V <sub>DPU5</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD6   | 17         | 14        | I/O   | Serial data 6. Connect to V <sub>DPU6</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC6   | 18         | 15        | I/O   | Serial clock 6. Connect to V <sub>DPU6</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SD7   | 19         | 16        | I/O   | Serial data 7. Connect to V <sub>DPU7</sub> <sup>(1)</sup> through a pull-up resistor                                          |
| SC7   | 20         | 17        | I/O   | Serial clock 7. Connect to V <sub>DPU7</sub> <sup>(1)</sup> through a pull-up resistor                                         |
| SCL   | 22         | 19        | I/O   | Serial clock bus. Connect to V <sub>DPU0</sub> <sup>(1)</sup> through a pull-up resistor                                       |
| SDA   | 23         | 20        | I/O   | Serial data bus. Connect to V <sub>DPU0</sub> <sup>(1)</sup> through a pull-up resistor                                        |
| VCC   | 24         | 21        | Power | Supply voltage                                                                                                                 |

(1) V<sub>DPUX</sub> is the pull-up reference voltage for the associated data line. V<sub>DPU0</sub> is the master I<sup>2</sup>C reference voltage and V<sub>DPU0</sub>–V<sub>DPU7</sub> are the slave channel reference voltages.

## 6 Specifications

### 6.1 Absolute Maximum Ratings<sup>(1)</sup>

over operating free-air temperature range (unless otherwise noted)

|                  |                              | MIN                     | MAX | UNIT |
|------------------|------------------------------|-------------------------|-----|------|
| V <sub>CC</sub>  | Supply voltage               | –0.5                    | 7   | V    |
| V <sub>I</sub>   | Input voltage <sup>(2)</sup> | –0.5                    | 7   | V    |
| I <sub>I</sub>   | Input current                | –20                     | 20  | mA   |
| I <sub>O</sub>   | Output current               | –25                     |     | mA   |
| I <sub>CC</sub>  | Supply current               | –100                    | 100 | mA   |
| T <sub>stg</sub> | Storage temperature          | –65                     | 150 | °C   |
| T <sub>J</sub>   | Max Junction Temperature     | V <sub>CC</sub> ≤ 3.6 V | 130 | °C   |
|                  |                              | V <sub>CC</sub> ≤ 5.5 V | 90  |      |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.

### 6.2 ESD Ratings

|                    |                         | VALUE                                                                          | UNIT  |
|--------------------|-------------------------|--------------------------------------------------------------------------------|-------|
| V <sub>(ESD)</sub> | Electrostatic discharge | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup>              | ±2000 |
|                    |                         | Charged-device model (CDM), per JEDEC specification JESD22-C101 <sup>(2)</sup> | ±1000 |

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 6.3 Recommended Operating Conditions

|                 |                                |                                  | MIN                   | MAX                   | UNIT |
|-----------------|--------------------------------|----------------------------------|-----------------------|-----------------------|------|
| V <sub>CC</sub> | Supply voltage                 | –40 °C ≤ T <sub>A</sub> ≤ 85 °C  | 1.65                  | 5.5                   | V    |
|                 |                                | 85 °C < T <sub>A</sub> ≤ 125 °C  | 1.65                  | 3.6                   |      |
| V <sub>IH</sub> | High-level input voltage       | SCL, SDA                         | 0.7 × V <sub>CC</sub> | 6                     | V    |
|                 |                                | A2–A0, $\overline{\text{RESET}}$ | 0.7 × V <sub>CC</sub> | V <sub>CC</sub> + 0.5 |      |
| V <sub>IL</sub> | Low-level input voltage        | SCL, SDA                         | –0.5                  | 0.3 × V <sub>CC</sub> | V    |
|                 |                                | A2–A0, $\overline{\text{RESET}}$ | –0.5                  | 0.3 × V <sub>CC</sub> |      |
| T <sub>A</sub>  | Operating free-air temperature | 3.6 V < V <sub>CC</sub> ≤ 5.5 V  | –40                   | 85                    | °C   |
|                 |                                | 1.65 V ≤ V <sub>CC</sub> ≤ 3.6 V | –40                   | 125                   |      |

### 6.4 Thermal Information

| THERMAL METRIC <sup>(1)</sup> |                                              | TCA9548A   |            | UNIT |
|-------------------------------|----------------------------------------------|------------|------------|------|
|                               |                                              | PW (TSSOP) | RGE (VQFN) |      |
|                               |                                              | 24 PINS    | 24 PINS    |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 108.8      | 57.2       | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 54.1       | 62.5       | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 62.7       | 34.4       | °C/W |
| ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 10.9       | 3.8        | °C/W |
| ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 62.3       | 34.4       | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | N/A        | 15.5       | °C/W |

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

## 6.5 Electrical Characteristics<sup>(1)</sup>

$V_{CC} = 1.65\text{ V to }5.5\text{ V}$ , over recommended operating free-air temperature ranges supported by [Recommended Operating Conditions](#) (unless otherwise noted)

| PARAMETER       |                                                         | TEST CONDITIONS                                              | $V_{CC}$         | MIN | TYP <sup>(2)</sup> | MAX  | UNIT          |
|-----------------|---------------------------------------------------------|--------------------------------------------------------------|------------------|-----|--------------------|------|---------------|
| $V_{PORR}$      | Power-on reset voltage, $V_{CC}$ rising                 | No load, $V_I = V_{CC}$ or GND <sup>(3)</sup>                |                  |     | 1.2                | 1.5  | V             |
| $V_{PORF}$      | Power-on reset voltage, $V_{CC}$ falling <sup>(4)</sup> | No load, $V_I = V_{CC}$ or GND <sup>(3)</sup>                |                  | 0.8 | 1                  |      | V             |
| $V_{O(SW)}$     | Switch output voltage                                   | $V_{I(SW)} = V_{CC}$ , $I_{SWout} = -100\text{ }\mu\text{A}$ | 5 V              |     | 3.6                |      | V             |
|                 |                                                         |                                                              | 4.5 V to 5.5 V   | 2.6 |                    | 4.5  |               |
|                 |                                                         |                                                              | 3.3 V            |     | 1.9                |      |               |
|                 |                                                         |                                                              | 3 V to 3.6 V     | 1.6 |                    | 2.8  |               |
|                 |                                                         |                                                              | 2.5 V            |     | 1.5                |      |               |
|                 |                                                         |                                                              | 2.3 V to 2.7 V   | 1.1 |                    | 2    |               |
|                 |                                                         |                                                              | 1.8 V            |     | 1.1                |      |               |
|                 |                                                         |                                                              | 1.65 V to 1.95 V | 0.6 |                    | 1.25 |               |
| $I_{OL}$        | SDA                                                     | $V_{OL} = 0.4\text{ V}$                                      | 1.65 V to 5.5 V  | 3   | 6                  |      | mA            |
|                 |                                                         | $V_{OL} = 0.6\text{ V}$                                      |                  | 6   | 9                  |      |               |
| $I_I$           | SCL, SDA                                                | $V_I = V_{CC}$ or GND <sup>(3)</sup>                         | 1.65 V to 5.5 V  | -1  |                    | 1    | $\mu\text{A}$ |
|                 | SC7–SC0, SD7–SD0                                        |                                                              |                  | -1  |                    | 1    |               |
|                 | A2–A0                                                   |                                                              |                  | -1  |                    | 1    |               |
|                 | RESET                                                   |                                                              |                  | -1  |                    | 1    |               |
| $I_{CC}$        | Operating mode                                          | $f_{SCL} = 400\text{ kHz}$                                   | 5.5 V            |     | 50                 | 80   | $\mu\text{A}$ |
|                 |                                                         |                                                              | 3.6 V            |     | 20                 | 35   |               |
|                 |                                                         |                                                              | 2.7 V            |     | 11                 | 20   |               |
|                 |                                                         |                                                              | 1.65 V           |     | 6                  | 10   |               |
|                 |                                                         | $f_{SCL} = 100\text{ kHz}$                                   | 5.5 V            |     | 9                  | 30   |               |
|                 |                                                         |                                                              | 3.6 V            |     | 6                  | 15   |               |
|                 |                                                         |                                                              | 2.7 V            |     | 4                  | 8    |               |
|                 |                                                         |                                                              | 1.65 V           |     | 2                  | 4    |               |
|                 | Standby mode                                            | Low inputs                                                   | 5.5 V            |     | 0.2                | 2    |               |
|                 |                                                         |                                                              | 3.6 V            |     | 0.1                | 2    |               |
|                 |                                                         |                                                              | 2.7 V            |     | 0.1                | 1    |               |
|                 |                                                         |                                                              | 1.65 V           |     | 0.1                | 1    |               |
|                 |                                                         | High inputs                                                  | 5.5 V            |     | 0.2                | 2    |               |
|                 |                                                         |                                                              | 3.6 V            |     | 0.1                | 2    |               |
|                 |                                                         |                                                              | 2.7 V            |     | 0.1                | 1    |               |
|                 |                                                         |                                                              | 1.65 V           |     | 0.1                | 1    |               |
|                 |                                                         | Low and High Inputs                                          | 3.6 V            |     | 1                  | 2    |               |
|                 |                                                         |                                                              | 2.7 V            |     | 0.7                | 1.5  |               |
|                 |                                                         |                                                              | 1.65 V           |     | 0.4                | 1    |               |
|                 |                                                         |                                                              |                  |     |                    |      |               |
| $\Delta I_{CC}$ | Supply-current change                                   | SCL, SDA                                                     | 1.65 V to 5.5 V  |     | 3                  | 20   | $\mu\text{A}$ |
|                 |                                                         |                                                              |                  |     | 3                  | 20   |               |
| $C_i$           | A2–A0                                                   | $V_I = V_{CC}$ or GND <sup>(3)</sup>                         | 1.65 V to 5.5 V  |     | 4                  | 5    | pF            |
|                 | RESET                                                   |                                                              |                  |     | 4                  | 5    |               |
|                 | SCL                                                     | $V_I = V_{CC}$ or GND <sup>(3)</sup> , Switch OFF            |                  |     | 20                 | 28   |               |

(1) For operation between specified voltage ranges, refer to the worst-case parameter in both applicable ranges.

(2) All typical values are at nominal supply voltage (1.8-, 2.5-, 3.3-, or 5-V  $V_{CC}$ ),  $T_A = 25^\circ\text{C}$ .

(3) RESET =  $V_{CC}$  (held high) when all other input voltages,  $V_I = \text{GND}$ .

(4) The power-on reset circuit resets the I<sup>2</sup>C bus logic with  $V_{CC} < V_{PORF}$ .

## Electrical Characteristics<sup>(1)</sup> (continued)

$V_{CC} = 1.65\text{ V to }5.5\text{ V}$ , over recommended operating free-air temperature ranges supported by [Recommended Operating Conditions](#) (unless otherwise noted)

| PARAMETER                    |                      | TEST CONDITIONS                                   | $V_{CC}$         | MIN | TYP <sup>(2)</sup> | MAX | UNIT     |
|------------------------------|----------------------|---------------------------------------------------|------------------|-----|--------------------|-----|----------|
| $C_{io(off)}$ <sup>(5)</sup> | SDA                  | $V_I = V_{CC}$ or GND <sup>(3)</sup> , Switch OFF | 1.65 V to 5.5 V  | 20  | 28                 |     | pF       |
|                              | SC7–SC0, SD7–SD0     |                                                   |                  | 5.5 | 7.5                |     |          |
| $R_{ON}$                     | Switch-on resistance | $V_O = 0.4\text{ V}$ , $I_O = 15\text{ mA}$       | 4.5 V to 5.5 V   | 4   | 10                 | 20  | $\Omega$ |
|                              |                      |                                                   | 3 V to 3.6 V     | 5   | 12                 | 30  |          |
|                              |                      | $V_O = 0.4\text{ V}$ , $I_O = 10\text{ mA}$       | 2.3 V to 2.7 V   | 7   | 15                 | 45  |          |
|                              |                      |                                                   | 1.65 V to 1.95 V | 10  | 25                 | 70  |          |

(5)  $C_{io(ON)}$  depends on internal capacitance and external capacitance added to the SCn lines when channel(s) are ON.

## 6.6 I<sup>2</sup>C Interface Timing Requirements

over recommended operating free-air temperature range (unless otherwise noted) (see [Figure 5](#))

|                      |                                                               |                                           | MIN               | MAX  | UNIT          |
|----------------------|---------------------------------------------------------------|-------------------------------------------|-------------------|------|---------------|
| <b>STANDARD MODE</b> |                                                               |                                           |                   |      |               |
| $f_{scl}$            | I <sup>2</sup> C clock frequency                              |                                           | 0                 | 100  | kHz           |
| $t_{sch}$            | I <sup>2</sup> C clock high time                              |                                           | 4                 |      | $\mu\text{s}$ |
| $t_{scl}$            | I <sup>2</sup> C clock low time                               |                                           | 4.7               |      | $\mu\text{s}$ |
| $t_{sp}$             | I <sup>2</sup> C spike time                                   |                                           |                   | 50   | ns            |
| $t_{sds}$            | I <sup>2</sup> C serial-data setup time                       |                                           | 250               |      | ns            |
| $t_{sdh}$            | I <sup>2</sup> C serial-data hold time                        |                                           | 0 <sup>(1)</sup>  |      | $\mu\text{s}$ |
| $t_{icr}$            | I <sup>2</sup> C input rise time                              |                                           |                   | 1000 | ns            |
| $t_{icf}$            | I <sup>2</sup> C input fall time                              |                                           |                   | 300  | ns            |
| $t_{ocf}$            | I <sup>2</sup> C output (SDn) fall time (10-pF to 400-pF bus) |                                           |                   | 300  | ns            |
| $t_{buf}$            | I <sup>2</sup> C bus free time between stop and start         |                                           | 4.7               |      | $\mu\text{s}$ |
| $t_{sts}$            | I <sup>2</sup> C start or repeated start condition setup      |                                           | 4.7               |      | $\mu\text{s}$ |
| $t_{sth}$            | I <sup>2</sup> C start or repeated start condition hold       |                                           | 4                 |      | $\mu\text{s}$ |
| $t_{sps}$            | I <sup>2</sup> C stop condition setup                         |                                           | 4                 |      | $\mu\text{s}$ |
| $t_{vdL(Data)}$      | Valid-data time (high to low) <sup>(2)</sup>                  | SCL low to SDA output low valid           |                   | 1    | $\mu\text{s}$ |
| $t_{vdH(Data)}$      | Valid-data time (low to high) <sup>(2)</sup>                  | SCL low to SDA output high valid          |                   | 0.6  | $\mu\text{s}$ |
| $t_{vd(ack)}$        | Valid-data time of ACK condition                              | ACK signal from SCL low to SDA output low |                   | 1    | $\mu\text{s}$ |
| $C_b$                | I <sup>2</sup> C bus capacitive load                          |                                           |                   | 400  | pF            |
| <b>FAST MODE</b>     |                                                               |                                           |                   |      |               |
| $f_{scl}$            | I <sup>2</sup> C clock frequency                              |                                           | 0                 | 400  | kHz           |
| $t_{sch}$            | I <sup>2</sup> C clock high time                              |                                           | 0.6               |      | $\mu\text{s}$ |
| $t_{scl}$            | I <sup>2</sup> C clock low time                               |                                           | 1.3               |      | $\mu\text{s}$ |
| $t_{sp}$             | I <sup>2</sup> C spike time                                   |                                           |                   | 50   | ns            |
| $t_{sds}$            | I <sup>2</sup> C serial-data setup time                       |                                           | 100               |      | ns            |
| $t_{sdh}$            | I <sup>2</sup> C serial-data hold time                        |                                           | 0 <sup>(1)</sup>  |      | $\mu\text{s}$ |
| $t_{icr}$            | I <sup>2</sup> C input rise time                              |                                           | $20 + 0.1C_{(3)}$ | 300  | ns            |
| $t_{icf}$            | I <sup>2</sup> C input fall time                              |                                           | $20 + 0.1C_{(3)}$ | 300  | ns            |
| $t_{ocf}$            | I <sup>2</sup> C output (SDn) fall time (10-pF to 400-pF bus) |                                           | $20 + 0.1C_{(3)}$ | 300  | ns            |

(1) A device internally must provide a hold time of at least 300 ns for the SDA signal (referred to the  $V_{IH}$  min of the SCL signal), to bridge the undefined region of the falling edge of SCL.

(2) Data taken using a 1-k $\Omega$  pull-up resistor and 50-pF load (see [Figure 6](#))

(3)  $C_b$  = total bus capacitance of one bus line in pF

## I<sup>2</sup>C Interface Timing Requirements (continued)

over recommended operating free-air temperature range (unless otherwise noted) (see [Figure 5](#))

|                        |                                                          |                                           | MIN | MAX | UNIT |
|------------------------|----------------------------------------------------------|-------------------------------------------|-----|-----|------|
| t <sub>buf</sub>       | I <sup>2</sup> C bus free time between stop and start    |                                           | 1.3 |     | μs   |
| t <sub>sts</sub>       | I <sup>2</sup> C start or repeated start condition setup |                                           | 0.6 |     | μs   |
| t <sub>sth</sub>       | I <sup>2</sup> C start or repeated start condition hold  |                                           | 0.6 |     | μs   |
| t <sub>sps</sub>       | I <sup>2</sup> C stop condition setup                    |                                           | 0.6 |     | μs   |
| t <sub>vdL(Data)</sub> | Valid-data time (high to low) <sup>(2)</sup>             | SCL low to SDA output low valid           |     | 1   | μs   |
| t <sub>vdH(Data)</sub> | Valid-data time (low to high) <sup>(2)</sup>             | SCL low to SDA output high valid          |     | 0.6 | μs   |
| t <sub>vd(ack)</sub>   | Valid-data time of ACK condition                         | ACK signal from SCL low to SDA output low |     | 1   | μs   |
| C <sub>b</sub>         | I <sup>2</sup> C bus capacitive load                     |                                           |     | 400 | pF   |

## 6.7 Reset Timing Requirements

over recommended operating free-air temperature range (unless otherwise noted)

| PARAMETER             |                                                       | MIN | MAX | UNIT |
|-----------------------|-------------------------------------------------------|-----|-----|------|
| t <sub>W(L)</sub>     | Pulse duration, $\overline{\text{RESET}}$ low         | 6   |     | ns   |
| t <sub>REC(STA)</sub> | Recovery time from $\overline{\text{RESET}}$ to start | 0   |     | ns   |

## 6.8 Switching Characteristics

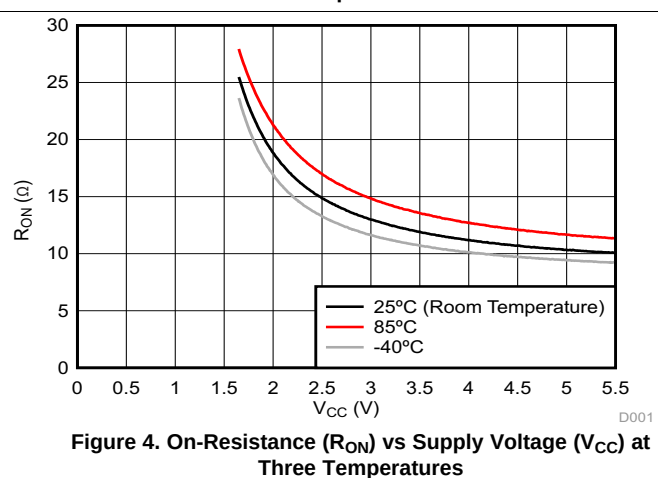
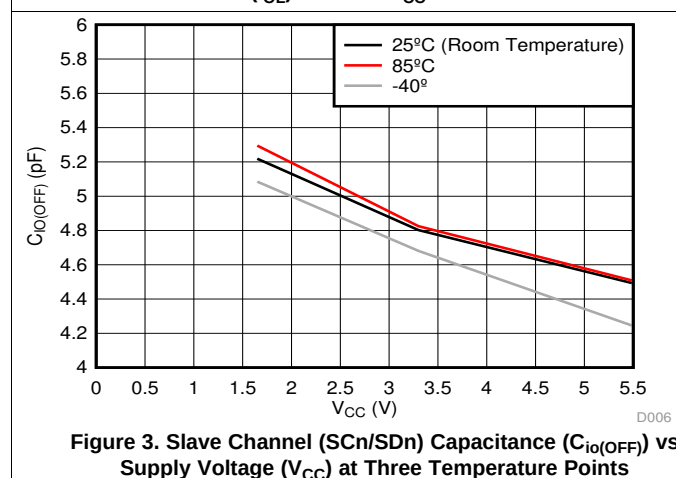
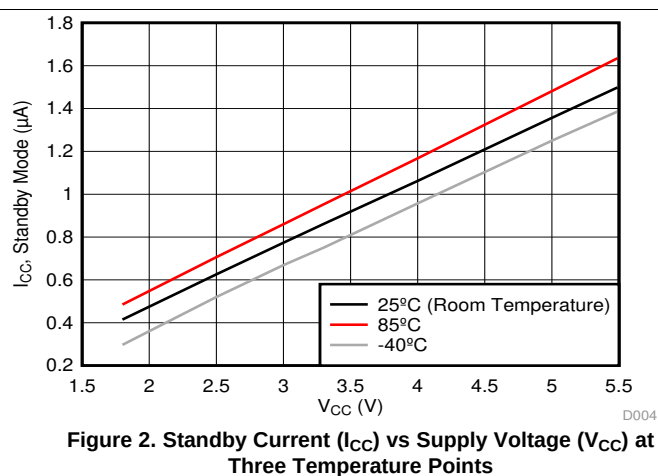
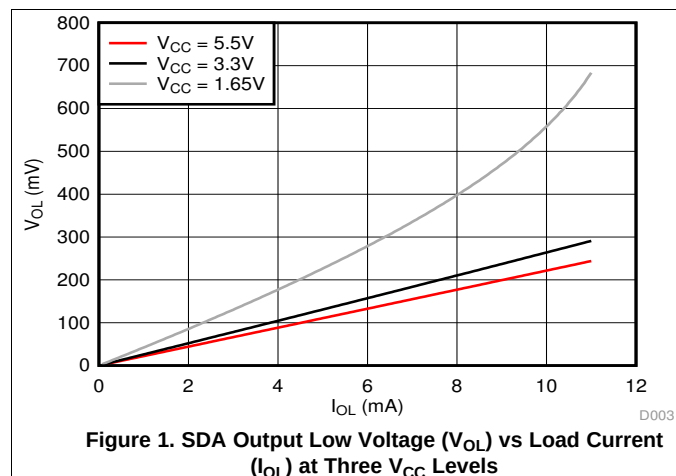
over recommended operating free-air temperature range, C<sub>L</sub> ≤ 100 pF (unless otherwise noted) (see [Figure 5](#))

| PARAMETER                       |                                            | FROM<br>(INPUT)                                | TO<br>(OUTPUT) | MIN | MAX | UNIT |
|---------------------------------|--------------------------------------------|------------------------------------------------|----------------|-----|-----|------|
| t <sub>pd</sub> <sup>(1)</sup>  | Propagation delay time                     | R <sub>ON</sub> = 20 Ω, C <sub>L</sub> = 15 pF | SDA or SCL     |     | 0.3 | ns   |
|                                 |                                            | R <sub>ON</sub> = 20 Ω, C <sub>L</sub> = 50 pF |                |     | 1   |      |
| t <sub>rst</sub> <sup>(2)</sup> | $\overline{\text{RESET}}$ time (SDA clear) | $\overline{\text{RESET}}$                      | SDA            |     | 500 | ns   |

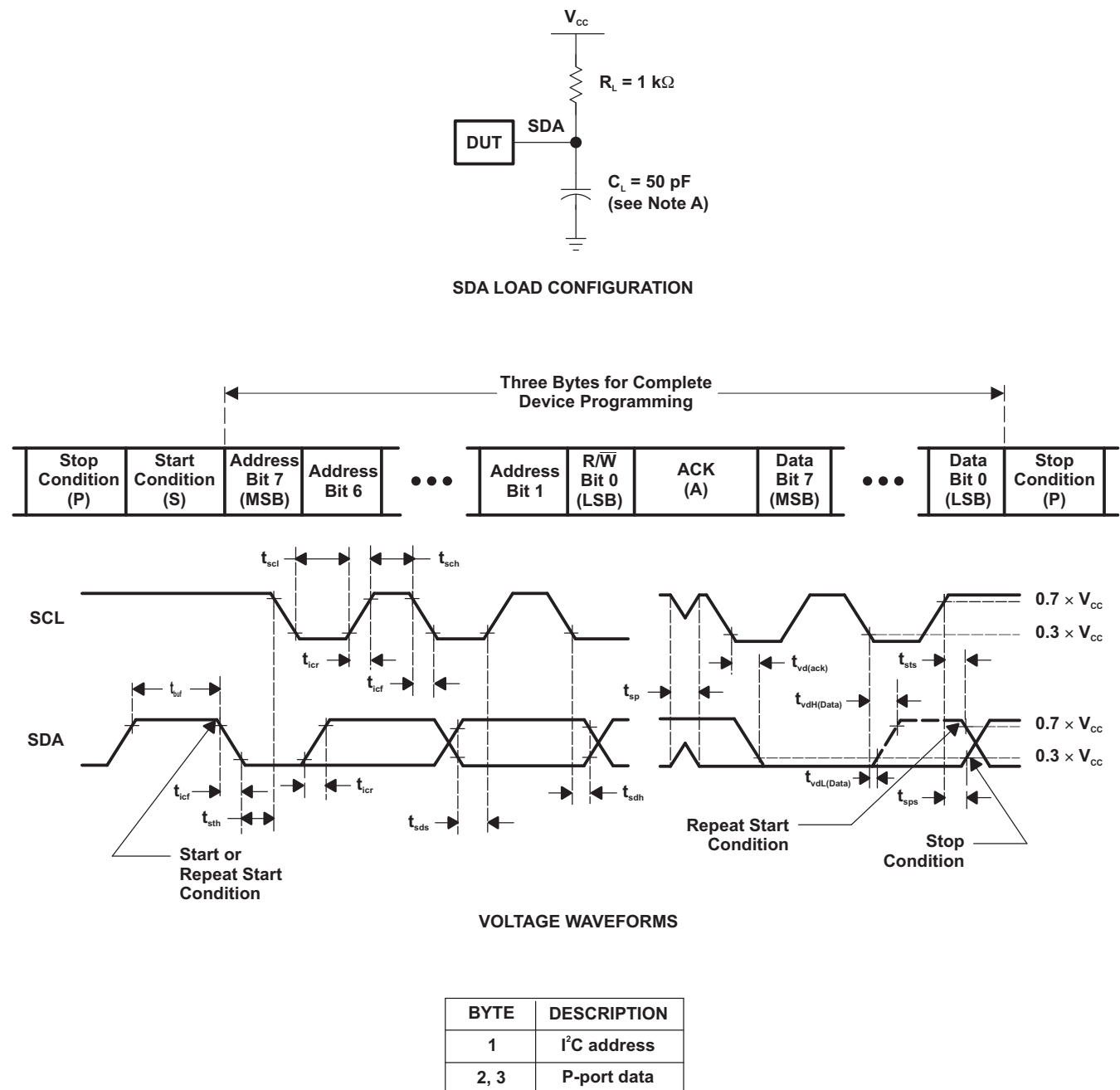
- (1) The propagation delay is the calculated RC time constant of the typical ON-state resistance of the switch and the specified load capacitance, when driven by an ideal voltage source (zero output impedance).
- (2) t<sub>rst</sub> is the propagation delay measured from the time the  $\overline{\text{RESET}}$  pin is first asserted low to the time the SDA pin is asserted high, signaling a stop condition. It must be a minimum of t<sub>WL</sub>.



## 6.9 Typical Characteristics



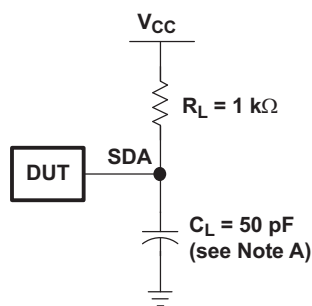
## 7 Parameter Measurement Information



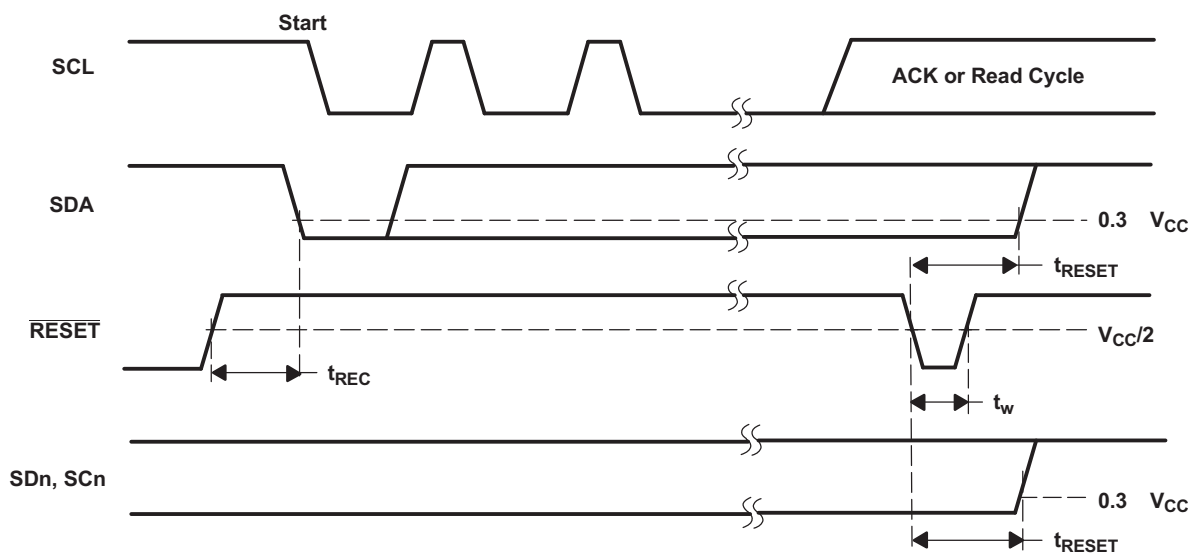
- A.  $C_L$  includes probe and jig capacitance.
- B. All inputs are supplied by generators having the following characteristics:  $PRR \leq 10 \text{ MHz}$ ,  $Z_O = 50 \Omega$ ,  $t_r/t_f \leq 30 \text{ ns}$ .
- C. Not all parameters and waveforms are applicable to all devices.

**Figure 5. I<sup>2</sup>C Load Circuit and Voltage Waveforms**

## Parameter Measurement Information (continued)



SDA LOAD CONFIGURATION



- A.  $C_L$  includes probe and jig capacitance.
- B. All inputs are supplied by generators having the following characteristics:  $PRR \leq 10\text{ MHz}$ ,  $Z_O = 50\text{ }\Omega$ ,  $t_r/t_f \leq 30\text{ ns}$ .
- C. I/Os are configured as inputs.
- D. Not all parameters and waveforms are applicable to all devices.

Figure 6. Reset Load Circuit and Voltage Waveforms

## 8 Detailed Description

### 8.1 Overview

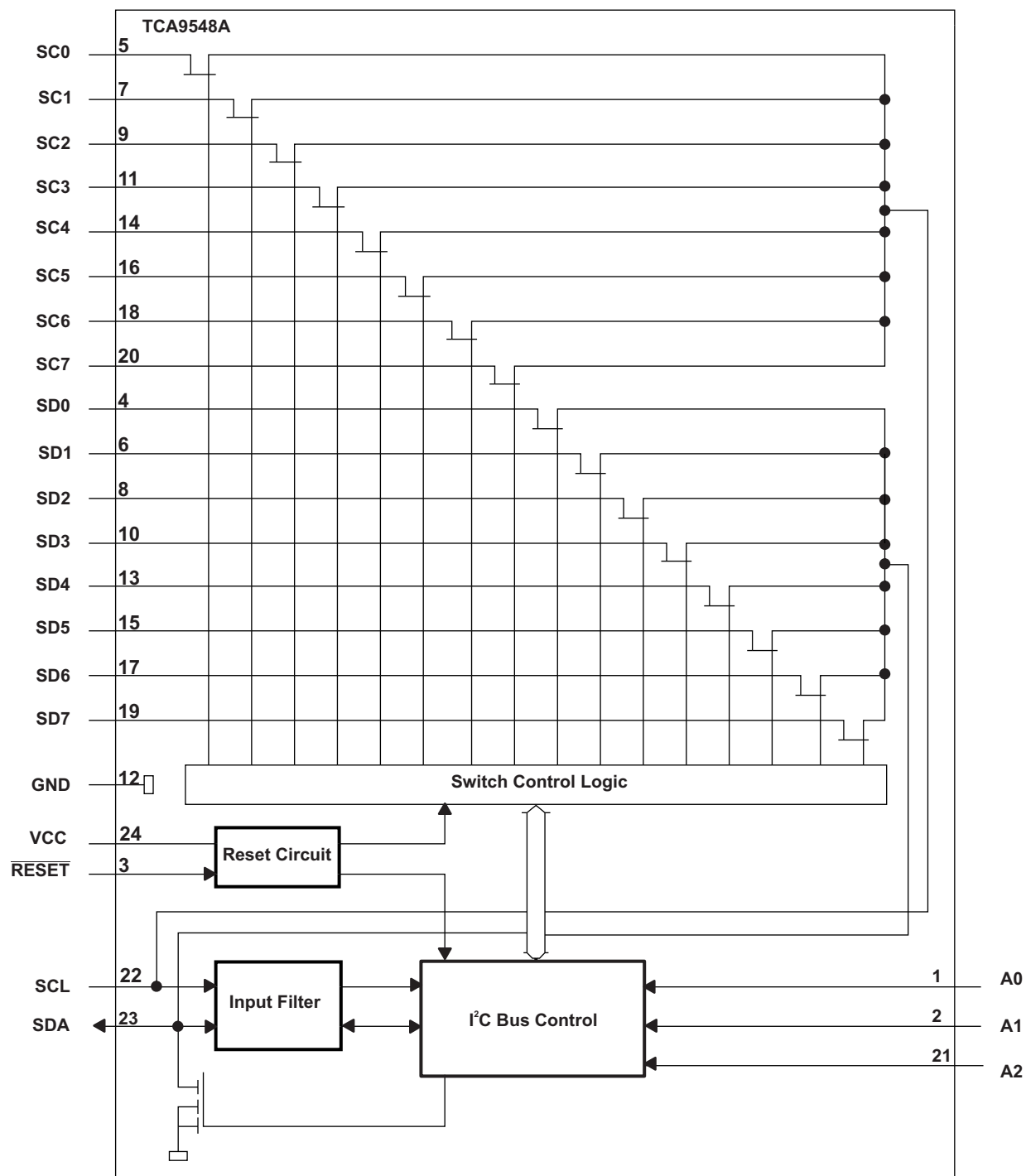
The TCA9548A is an 8-channel, bidirectional translating I<sup>2</sup>C switch. The master SCL/SDA signal pair is directed to eight channels of slave devices, SC0/SD0-SC7/SD7. Any individual downstream channel can be selected as well as any combination of the eight channels.

The device offers an active-low  $\overline{\text{RESET}}$  input which resets the state machine and allows the TCA9548A to recover must one of the downstream I<sup>2</sup>C buses get stuck in a low state. The state machine of the device can also be reset by cycling the power supply,  $V_{CC}$ , also known as a power-on reset (POR). Both the  $\overline{\text{RESET}}$  function and a POR cause all channels to be deselected.

The connections of the I<sup>2</sup>C data path are controlled by the same I<sup>2</sup>C master device that is switched to communicate with multiple I<sup>2</sup>C slaves. After the successful acknowledgment of the slave address (hardware selectable by A0, A1, and A2 pins), a single 8-bit control register is written to or read from to determine the selected channels.

The TCA9548A may also be used for voltage translation, allowing the use of different bus voltages on each SCn/SDn pair such that 1.8-V, 2.5-V, or 3.3-V parts can communicate with 5-V parts. This is achieved by using external pull-up resistors to pull the bus up to the desired voltage for the master and each slave channel.

## 8.2 Functional Block Diagram



### 8.3 Feature Description

The TCA9548A is an 8-channel, bidirectional translating switch for I<sup>2</sup>C buses that supports Standard-Mode (100 kHz) and Fast-Mode (400 kHz) operation. The TCA9548A features I<sup>2</sup>C control using a single 8-bit control register in which each bit controls the enabling and disabling of one of the corresponding 8 switch channels for I<sup>2</sup>C data flow. Depending on the application, voltage translation of the I<sup>2</sup>C bus can also be achieved using the TCA9548A to allow 1.8-V, 2.5-V, or 3.3-V parts to communicate with 5-V parts. Additionally, in the event that communication on the I<sup>2</sup>C bus enters a fault state, the TCA9548A can be reset to resume normal operation using the  $\overline{\text{RESET}}$  pin feature or by a power-on reset which results from cycling power to the device.

### 8.4 Device Functional Modes

#### 8.4.1 $\overline{\text{RESET}}$ Input

The  $\overline{\text{RESET}}$  input is an active-low signal that may be used to recover from a bus-fault condition. When this signal is asserted low for a minimum of  $t_{\text{WL}}$ , the TCA9548A resets its registers and I<sup>2</sup>C state machine and deselects all channels. The  $\overline{\text{RESET}}$  input must be connected to V<sub>CC</sub> through a pull-up resistor.

#### 8.4.2 Power-On Reset

When power is applied to the VCC pin, an internal power-on reset holds the TCA9548A in a reset condition until V<sub>CC</sub> has reached V<sub>PORR</sub>. At this point, the reset condition is released, and the TCA9548A registers and I<sup>2</sup>C state machine are initialized to their default states, all zeroes, causing all the channels to be deselected. Thereafter, V<sub>CC</sub> must be lowered below V<sub>PORF</sub> to reset the device.

### 8.5 Programming

#### 8.5.1 I<sup>2</sup>C Interface

The TCA9548A has a standard bidirectional I<sup>2</sup>C interface that is controlled by a master device in order to be configured or read the status of this device. Each slave on the I<sup>2</sup>C bus has a specific device address to differentiate between other slave devices that are on the same I<sup>2</sup>C bus. Many slave devices require configuration upon startup to set the behavior of the device. This is typically done when the master accesses internal register maps of the slave, which have unique register addresses. A device can have one or multiple registers where data is stored, written, or read.

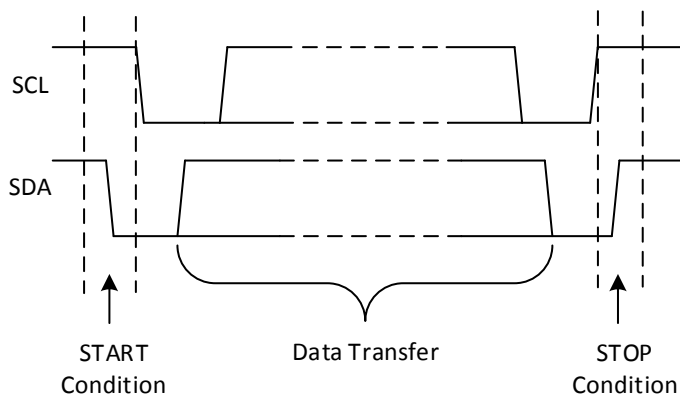
The physical I<sup>2</sup>C interface consists of the serial clock (SCL) and serial data (SDA) lines. Both SDA and SCL lines must be connected to V<sub>CC</sub> through a pull-up resistor. The size of the pull-up resistor is determined by the amount of capacitance on the I<sup>2</sup>C lines. (For further details, see the [I<sup>2</sup>C Pull-up Resistor Calculation](#) application report. Data transfer may be initiated only when the bus is idle. A bus is considered idle if both SDA and SCL lines are high after a STOP condition (See [Figure 7](#) and [Figure 8](#)).

The following is the general procedure for a master to access a slave device:

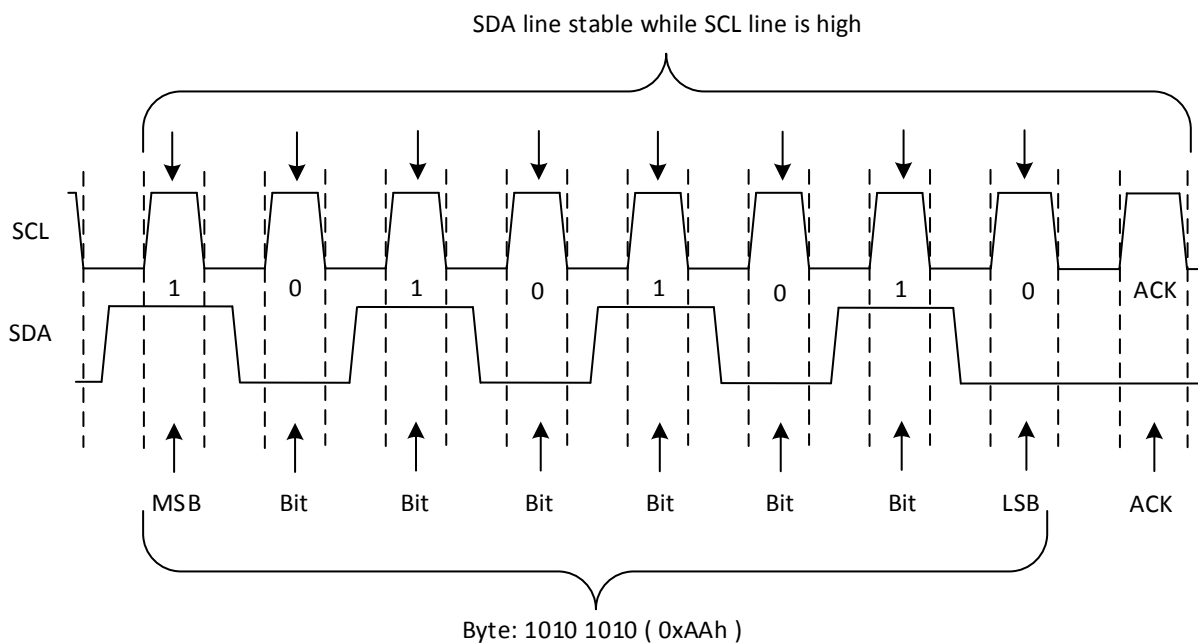
1. If a master wants to send data to a slave:
  - Master-transmitter sends a START condition and addresses the slave-receiver.
  - Master-transmitter sends data to slave-receiver.
  - Master-transmitter terminates the transfer with a STOP condition.
2. If a master wants to receive or read data from a slave:
  - Master-receiver sends a START condition and addresses the slave-transmitter.
  - Master-receiver sends the requested register to read to slave-transmitter.
  - Master-receiver receives data from the slave-transmitter.

## Programming (continued)

- Master-receiver terminates the transfer with a STOP condition.



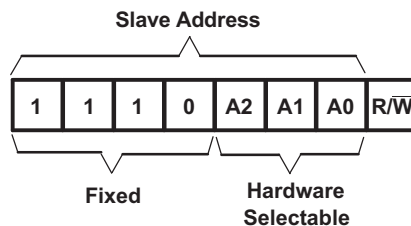
**Figure 7. Definition of Start and Stop Conditions**



**Figure 8. Bit Transfer**

### 8.5.2 Device Address

Figure 9 shows the address byte of the TCA9548A.



**Figure 9. TCA9548A Address**

## Programming (continued)

The last bit of the slave address defines the operation (read or write) to be performed. When it is high (1), a read is selected, while a low (0) selects a write operation.

Table 1 shows the TCA9548A address reference.

**Table 1. Address Reference**

| INPUTS |    |    | I <sup>2</sup> C BUS SLAVE ADDRESS |
|--------|----|----|------------------------------------|
| A2     | A1 | A0 |                                    |
| L      | L  | L  | 112 (decimal), 70 (hexadecimal)    |
| L      | L  | H  | 113 (decimal), 71 (hexadecimal)    |
| L      | H  | L  | 114 (decimal), 72 (hexadecimal)    |
| L      | H  | H  | 115 (decimal), 73 (hexadecimal)    |
| H      | L  | L  | 116 (decimal), 74 (hexadecimal)    |
| H      | L  | H  | 117 (decimal), 75 (hexadecimal)    |
| H      | H  | L  | 118 (decimal), 76 (hexadecimal)    |
| H      | H  | H  | 119 (decimal), 77 (hexadecimal)    |

### 8.5.3 Bus Transactions

Data must be sent to and received from the slave devices, and this is accomplished by reading from or writing to registers in the slave device.

Registers are locations in the memory of the slave which contain information, whether it be the configuration information or some sampled data to send back to the master. The master must write information to these registers in order to instruct the slave device to perform a task.

While it is common to have registers in I<sup>2</sup>C slaves, note that not all slave devices have registers. Some devices are simple and contain only 1 register, which may be written to directly by sending the register data immediately after the slave address, instead of addressing a register. The TCA9548A is example of a single-register device, which is controlled via I<sup>2</sup>C commands. Since it has 1 bit to enable or disable a channel, there is only 1 register needed, and the master merely writes the register data after the slave address, skipping the register number.

#### 8.5.3.1 Writes

To write on the I<sup>2</sup>C bus, the master sends a START condition on the bus with the address of the slave, as well as the last bit (the R/W bit) set to 0, which signifies a write. The slave acknowledges, letting the master know it is ready. After this, the master starts sending the control register data to the slave until the master has sent all the data necessary (which is sometimes only a single byte), and the master terminates the transmission with a STOP condition.

There is no limit to the number of bytes sent, but the last byte sent is what is in the register.

Figure 10 shows an example of writing a single byte to a slave register.



- Master controls SDA line
- Slave controls SDA line

### Write to one register in a device

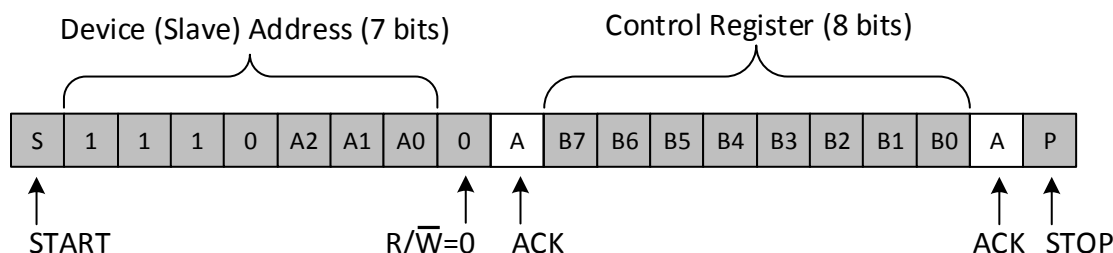


Figure 10. Write to Register

#### 8.5.3.2 Reads

Reading from a slave is very similar to writing, but the master sends a START condition, followed by the slave address with the R/W bit set to 1 (signifying a read). The slave acknowledges the read request, and the master releases the SDA bus but continues supplying the clock to the slave. During this part of the transaction, the master becomes the master-receiver, and the slave becomes the slave-transmitter.

The master continues to send out the clock pulses, but releases the SDA line so that the slave can transmit data. At the end of every byte of data, the master sends an ACK to the slave, letting the slave know that it is ready for more data. Once the master has received the number of bytes it is expecting, it sends a NACK, signaling to the slave to halt communications and release the bus. The master follows this up with a STOP condition.

Figure 11 shows an example of reading a single byte from a slave register.

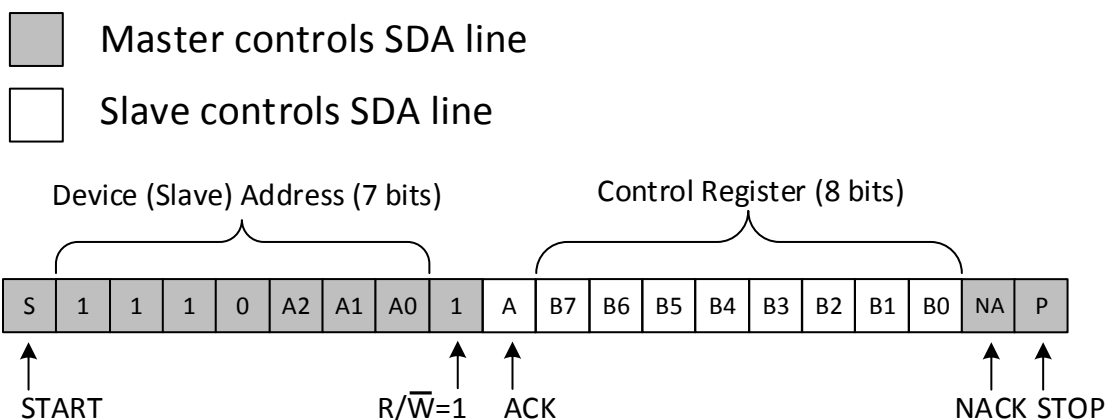
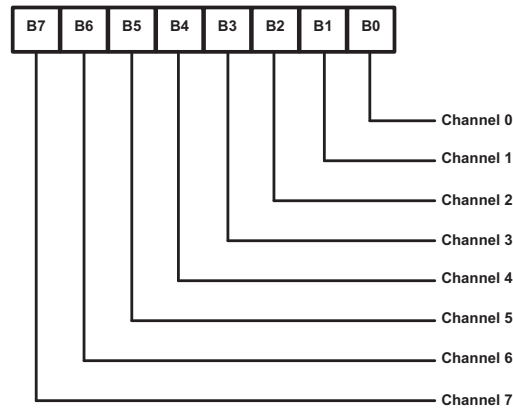


Figure 11. Read from Control Register

### 8.5.4 Control Register

Following the successful acknowledgment of the address byte, the bus master sends a command byte that is stored in the control register in the TCA9548A (see [Figure 12](#)). This register can be written and read via the I<sup>2</sup>C bus. Each bit in the command byte corresponds to a SCn/SDn channel and a high (or 1) selects this channel. Multiple SCn/SDn channels may be selected at the same time. When a channel is selected, the channel becomes active after a stop condition has been placed on the I<sup>2</sup>C bus. This ensures that all SCn/SDn lines are in a high state when the channel is made active, so that no false conditions are generated at the time of connection. A stop condition always must occur immediately after the acknowledge cycle. If multiple bytes are received by the TCA9548A, it saves the last byte received.

Channel Selection Bits (Read/Write)



**Figure 12. Control Register**

[Table 2](#) shows the TCA9548A Command Byte Definition.

**Table 2. Command Byte Definition**

| CONTROL REGISTER BITS |    |    |    |    |    |    |    | COMMAND                                           |
|-----------------------|----|----|----|----|----|----|----|---------------------------------------------------|
| B7                    | B6 | B5 | B4 | B3 | B2 | B1 | B0 |                                                   |
| X                     | X  | X  | X  | X  | X  | X  | 0  | Channel 0 disabled                                |
|                       |    |    |    |    |    |    | 1  | Channel 0 enabled                                 |
| X                     | X  | X  | X  | X  | X  | 0  | X  | Channel 1 disabled                                |
|                       |    |    |    |    |    | 1  |    | Channel 1 enabled                                 |
| X                     | X  | X  | X  | X  | 0  | X  | X  | Channel 2 disabled                                |
|                       |    |    |    |    | 1  |    |    | Channel 2 enabled                                 |
| X                     | X  | X  | X  | 0  | X  | X  | X  | Channel 3 disabled                                |
|                       |    |    |    | 1  |    |    |    | Channel 3 enabled                                 |
| X                     | X  | X  | 0  | X  | X  | X  | X  | Channel 4 disabled                                |
|                       |    |    | 1  |    |    |    |    | Channel 4 enabled                                 |
| X                     | X  | 0  | X  | X  | X  | X  | X  | Channel 5 disabled                                |
|                       |    | 1  |    |    |    |    |    | Channel 5 enabled                                 |
| X                     | 0  | X  | X  | X  | X  | X  | X  | Channel 6 disabled                                |
|                       | 1  |    |    |    |    |    |    | Channel 6 enabled                                 |
| 0                     | X  | X  | X  | X  | X  | X  | X  | Channel 7 disabled                                |
| 1                     |    |    |    |    |    |    |    | Channel 7 enabled                                 |
| 0                     | 0  | 0  | 0  | 0  | 0  | 0  | 0  | No channel selected, power-up/reset default state |

### 8.5.5 $\overline{\text{RESET}}$ Input

The  $\overline{\text{RESET}}$  input is an active-low signal that may be used to recover from a bus-fault condition. When this signal is asserted low for a minimum of  $t_{\text{WL}}$ , the TCA9548A resets its registers and I<sup>2</sup>C state machine and deselects all channels. The  $\overline{\text{RESET}}$  input must be connected to  $V_{\text{CC}}$  through a pull-up resistor.

### 8.5.6 Power-On Reset

When power (from 0 V) is applied to  $V_{\text{CC}}$ , an internal power-on reset holds the TCA9548A in a reset condition until  $V_{\text{CC}}$  has reached  $V_{\text{POR}}$ . At that point, the reset condition is released and the TCA9548A registers and I<sup>2</sup>C state machine initialize to their default states. After that,  $V_{\text{CC}}$  must be lowered to below  $V_{\text{POR}}$  and then back up to the operating voltage for a power-reset cycle.

## 9 Application and Implementation

### NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

### 9.1 Application Information

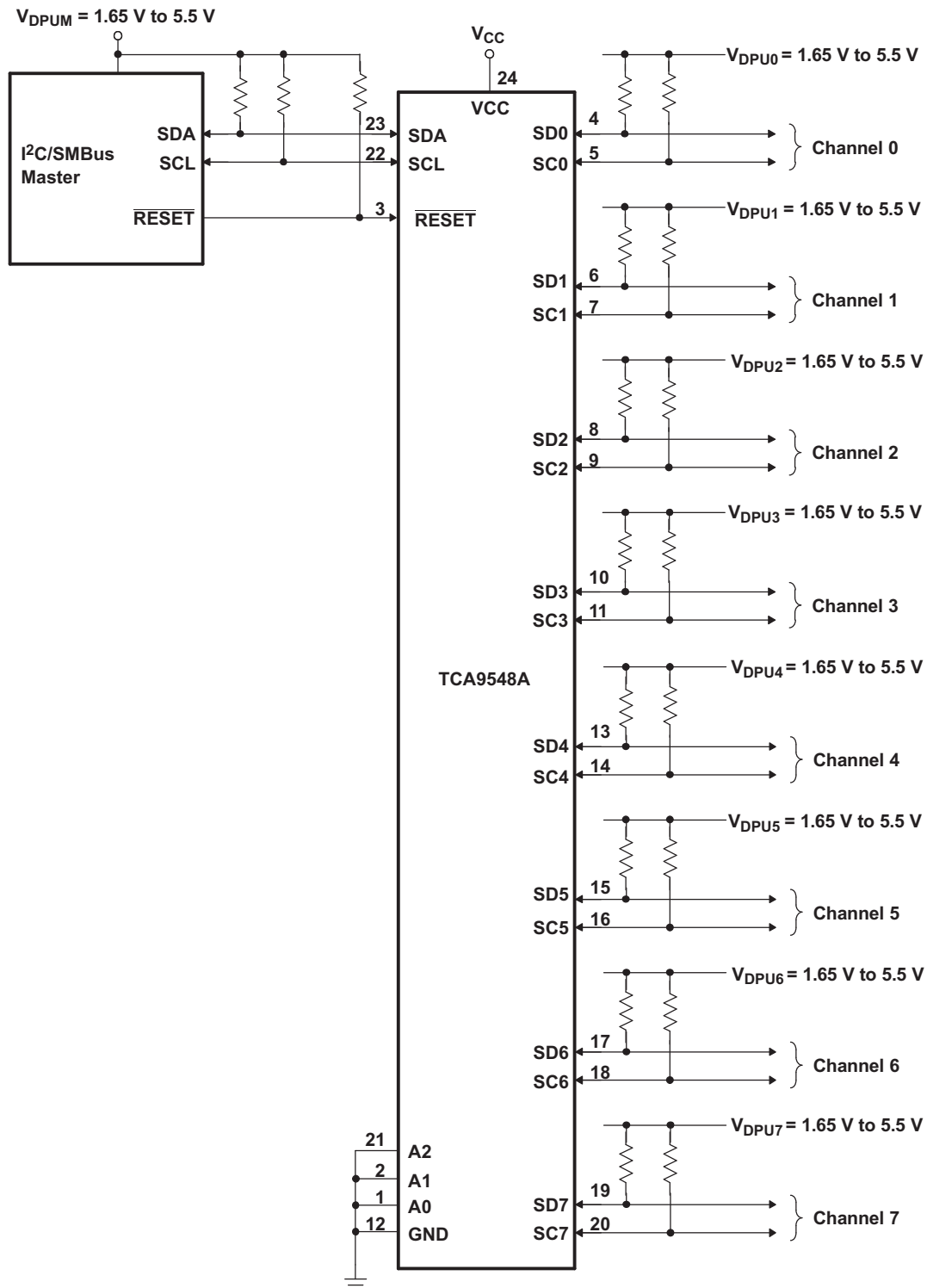
Applications of the TCA9548A contain an I<sup>2</sup>C (or SMBus) master device and up to eight I<sup>2</sup>C slave devices. The downstream channels are ideally used to resolve I<sup>2</sup>C slave address conflicts. For example, if eight identical digital temperature sensors are needed in the application, one sensor can be connected at each channel: 0-7. When the temperature at a specific location needs to be read, the appropriate channel can be enabled and all other channels switched off, the data can be retrieved, and the I<sup>2</sup>C master can move on and read the next channel.

In an application where the I<sup>2</sup>C bus contains many additional slave devices that do not result in I<sup>2</sup>C slave address conflicts, these slave devices can be connected to any desired channel to distribute the total bus capacitance across multiple channels. If multiple switches are enabled simultaneously, additional design requirements must be considered (see the [Design Requirements](#) section and [Detailed Design Procedure](#) section).

### 9.2 Typical Application

[Figure 13](#) shows an application in which the TCA9548A can be used.

## Typical Application (continued)



Pin numbers shown are for the PW package.

**Figure 13. Typical Application Schematic**

## Typical Application (continued)

### 9.2.1 Design Requirements

A typical application of the TCA9548A contains one or more data pull-up voltages,  $V_{DPUX}$ , one for the master device ( $V_{DPUM}$ ) and one for each of the selectable slave channels ( $V_{DPU0} - V_{DPU7}$ ). In the event where the master device and all slave devices operate at the same voltage, then  $V_{DPUM} = V_{DPUX} = V_{CC}$ . In an application where voltage translation is necessary, additional design requirements must be considered to determine an appropriate  $V_{CC}$  voltage.

The A0, A1, and A2 pins are hardware selectable to control the slave address of the TCA9548A. These pins may be tied directly to GND or  $V_{CC}$  in the application.

If multiple slave channels are activated simultaneously in the application, then the total  $I_{OL}$  from SCL/SDA to GND on the master side is the sum of the currents through all pull-up resistors,  $R_p$ .

The pass-gate transistors of the TCA9548A are constructed such that the  $V_{CC}$  voltage can be used to limit the maximum voltage that is passed from one I<sup>2</sup>C bus to another.

Figure 14 shows the voltage characteristics of the pass-gate transistors (note that the graph was generated using data specified in the [Electrical Characteristics](#) table). In order for the TCA9548A to act as a voltage translator, the  $V_{pass}$  voltage must be equal to or lower than the lowest bus voltage. For example, if the main bus is running at 5 V and the downstream buses are 3.3 V and 2.7 V,  $V_{pass}$  must be equal to or below 2.7 V to effectively clamp the downstream bus voltages. As shown in Figure 14,  $V_{pass(max)}$  is 2.7 V when the TCA9548A supply voltage is 4 V or lower, so the TCA9548A supply voltage could be set to 3.3 V. Pull-up resistors then can be used to bring the bus voltages to their appropriate levels (see Figure 13).

### 9.2.2 Detailed Design Procedure

Once all the slaves are assigned to the appropriate slave channels and bus voltages are identified, the pull-up resistors,  $R_p$ , for each of the buses need to be selected appropriately. The minimum pull-up resistance is a function of  $V_{DPUX}$ ,  $V_{OL(max)}$ , and  $I_{OL}$  as shown in Equation 1:

$$R_{p(min)} = \frac{V_{DPUX} - V_{OL(max)}}{I_{OL}} \quad (1)$$

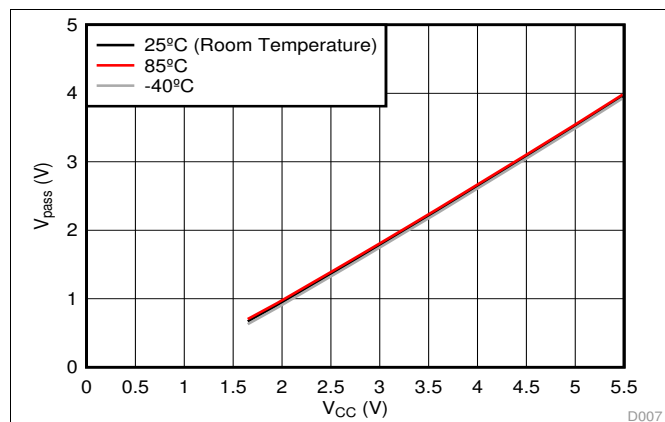
The maximum pull-up resistance is a function of the maximum rise time,  $t_r$  (300 ns for fast-mode operation,  $f_{SCL} = 400$  kHz) and bus capacitance,  $C_b$  as shown in Equation 2:

$$R_{p(max)} = \frac{t_r}{0.8473 \times C_b} \quad (2)$$

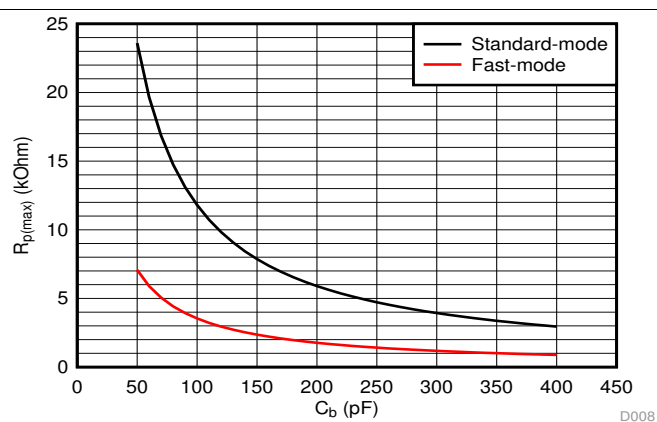
The maximum bus capacitance for an I<sup>2</sup>C bus must not exceed 400 pF for fast-mode operation. The bus capacitance can be approximated by adding the capacitance of the TCA9548A,  $C_{io(OFF)}$ , the capacitance of wires, connections and traces, and the capacitance of each individual slave on a given channel. If multiple channels are activated simultaneously, each of the slaves on all channels contribute to total bus capacitance.

## Typical Application (continued)

### 9.2.3 Application Curves

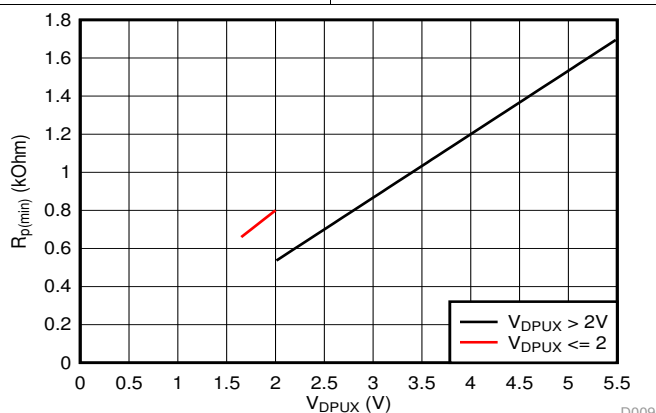


**Figure 14. Pass-Gate Voltage ( $V_{pass}$ ) vs Supply Voltage ( $V_{CC}$ ) at Three Temperature Points**



Standard-mode ( $f_{SCL} = 100$  kHz,  $t_r = 1$   $\mu$ s)      Fast-mode ( $f_{SCL} = 400$  kHz,  $t_r = 300$  ns)

**Figure 15. Maximum Pull-Up Resistance ( $R_{p(max)}$ ) vs Bus Capacitance ( $C_b$ )**



$V_{OL} = 0.2 \cdot V_{DPUX}$ ,  $I_{OL} = 2$  mA when  $V_{DPUX} \leq 2$  V  
 $V_{OL} = 0.4$  V,  $I_{OL} = 3$  mA when  $V_{DPUX} > 2$  V

**Figure 16. Minimum Pullup Resistance ( $R_{p(min)}$ ) vs Pullup Reference Voltage ( $V_{DPUX}$ )**

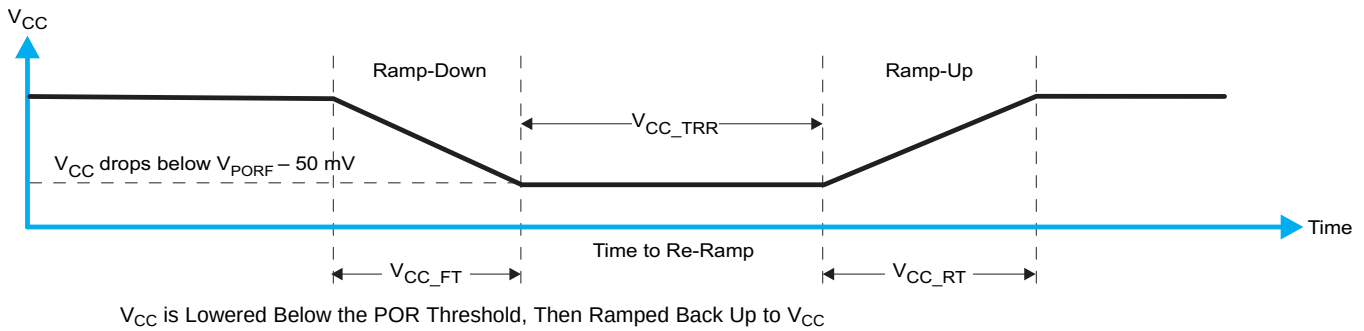
## 10 Power Supply Recommendations

The operating power-supply voltage range of the TCA9548A is 1.65 V to 5.5 V applied at the VCC pin. When the TCA9548A is powered on for the first time or anytime the device must be reset by cycling the power supply, the power-on reset requirements must be followed to ensure the I<sup>2</sup>C bus logic is initialized properly.

### 10.1 Power-On Reset Requirements

In the event of a glitch or data corruption, TCA9548A can be reset to its default conditions by using the power-on reset feature. Power-on reset requires that the device go through a power cycle to be completely reset. This reset also happens when the device is powered on for the first time in an application.

A power-on reset is shown in Figure 17.



**Figure 17. Power-On Reset Waveform**

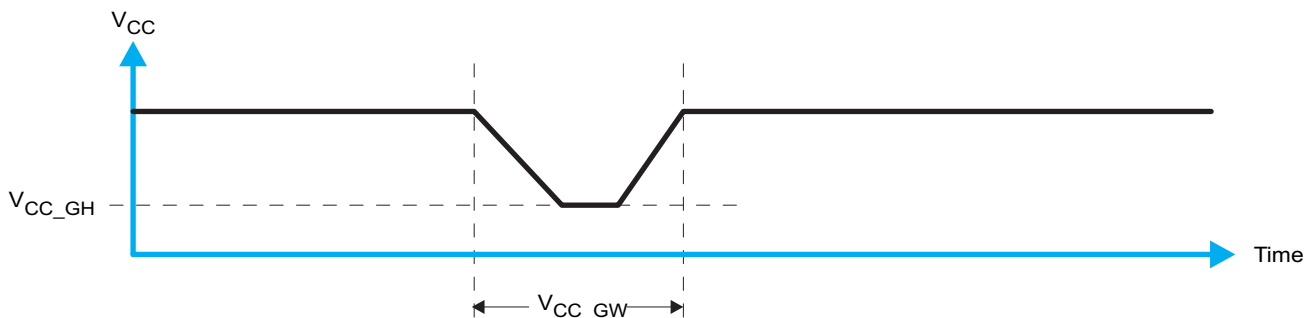
Table 3 specifies the performance of the power-on reset feature for TCA9548A for both types of power-on reset.

**Table 3. Recommended Supply Sequencing and Ramp Rates<sup>(1)</sup>**

| PARAMETER           |                                                                                                                        |               | MIN | MAX | UNIT |
|---------------------|------------------------------------------------------------------------------------------------------------------------|---------------|-----|-----|------|
| V <sub>CC_FT</sub>  | Fall time                                                                                                              | See Figure 17 | 1   | 100 | ms   |
| V <sub>CC_RT</sub>  | Rise time                                                                                                              | See Figure 17 | 0.1 | 100 | ms   |
| V <sub>CC_TRR</sub> | Time to re-ramp (when V <sub>CC</sub> drops below V <sub>PORF(min)</sub> – 50 mV or when V <sub>CC</sub> drops to GND) | See Figure 17 | 40  |     | μs   |
| V <sub>CC_GH</sub>  | Level that V <sub>CC</sub> can glitch down to, but not cause a functional disruption when V <sub>CC_GW</sub> = 1 μs    | See Figure 18 |     | 1.2 | V    |
| V <sub>CC_GW</sub>  | Glitch width that does not cause a functional disruption when V <sub>CC_GH</sub> = 0.5 × V <sub>CC</sub>               | See Figure 18 |     | 10  | μs   |

(1) All supply sequencing and ramp rate values are measured at T<sub>A</sub> = 25°C

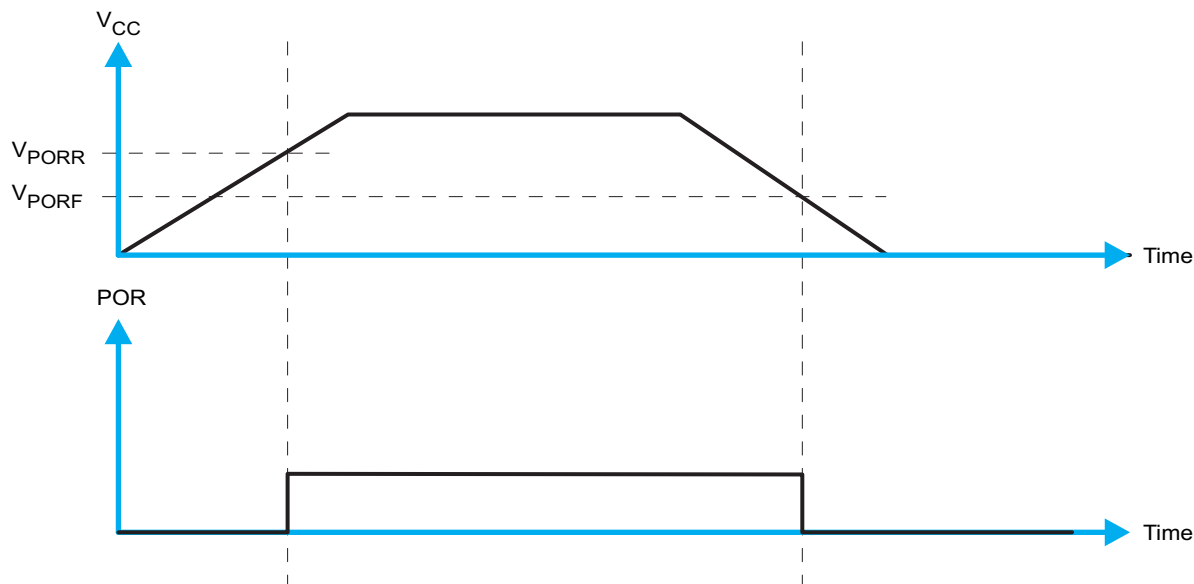
Glitches in the power supply can also affect the power-on reset performance of this device. The glitch width (V<sub>CC\_GW</sub>) and height (V<sub>CC\_GH</sub>) are dependent on each other. The bypass capacitance, source impedance, and device impedance are factors that affect power-on reset performance. Figure 18 and Table 3 provide more information on how to measure these specifications.



**Figure 18. Glitch Width and Glitch Height**



$V_{POR}$  is critical to the power-on reset.  $V_{POR}$  is the voltage level at which the reset condition is released and all the registers and the I<sup>2</sup>C/SMBus state machine are initialized to their default states. The value of  $V_{POR}$  differs based on the  $V_{CC}$  being lowered to or from 0. [Figure 19](#) and [Table 3](#) provide more details on this specification.



**Figure 19.  $V_{POR}$**



## 12 Device and Documentation Support

### 12.1 Documentation Support

#### 12.1.1 Related Documentation

For related documentation see the following:

- [I2C Bus Pull-Up Resistor Calculation](#)
- [Maximum Clock Frequency of I2C Bus Using Repeaters](#)
- [Introduction to Logic](#)
- [Understanding the I2C Bus](#)
- [Choosing the Correct I2C Device for New Designs](#)
- [TCA9548AEVM User's Guide](#)

### 12.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on [ti.com](http://ti.com). In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

### 12.3 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

### 12.4 Trademarks

E2E is a trademark of Texas Instruments.

All other trademarks are the property of their respective owners.

### 12.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

### 12.6 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

## 13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

## PACKAGING INFORMATION

| Orderable Device | Status<br>(1) | Package Type | Package<br>Drawing | Pins | Package<br>Qty | Eco Plan<br>(2)            | Lead/Ball Finish<br>(6) | MSL Peak Temp<br>(3) | Op Temp (°C) | Device Marking<br>(4/5) | Samples                 |
|------------------|---------------|--------------|--------------------|------|----------------|----------------------------|-------------------------|----------------------|--------------|-------------------------|-------------------------|
| TCA9548AMRGER    | ACTIVE        | VQFN         | RGE                | 24   | 3000           | Green (RoHS<br>& no Sb/Br) | NIPDAU                  | Level-2-260C-1 YEAR  | -40 to 85    | PW548A                  | <a href="#">Samples</a> |
| TCA9548APWR      | ACTIVE        | TSSOP        | PW                 | 24   | 2000           | Green (RoHS<br>& no Sb/Br) | NIPDAU                  | Level-1-260C-UNLIM   | -40 to 85    | PW548A                  | <a href="#">Samples</a> |
| TCA9548ARGER     | ACTIVE        | VQFN         | RGE                | 24   | 3000           | Green (RoHS<br>& no Sb/Br) | NIPDAU                  | Level-2-260C-1 YEAR  | -40 to 85    | PW548A                  | <a href="#">Samples</a> |

(1) The marketing status values are defined as follows:

**ACTIVE:** Product device recommended for new designs.

**LIFEBUY:** TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

**NRND:** Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

**PREVIEW:** Device has been announced but is not in production. Samples may or may not be available.

**OBSOLETE:** TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

**RoHS Exempt:** TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

**Green:** TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

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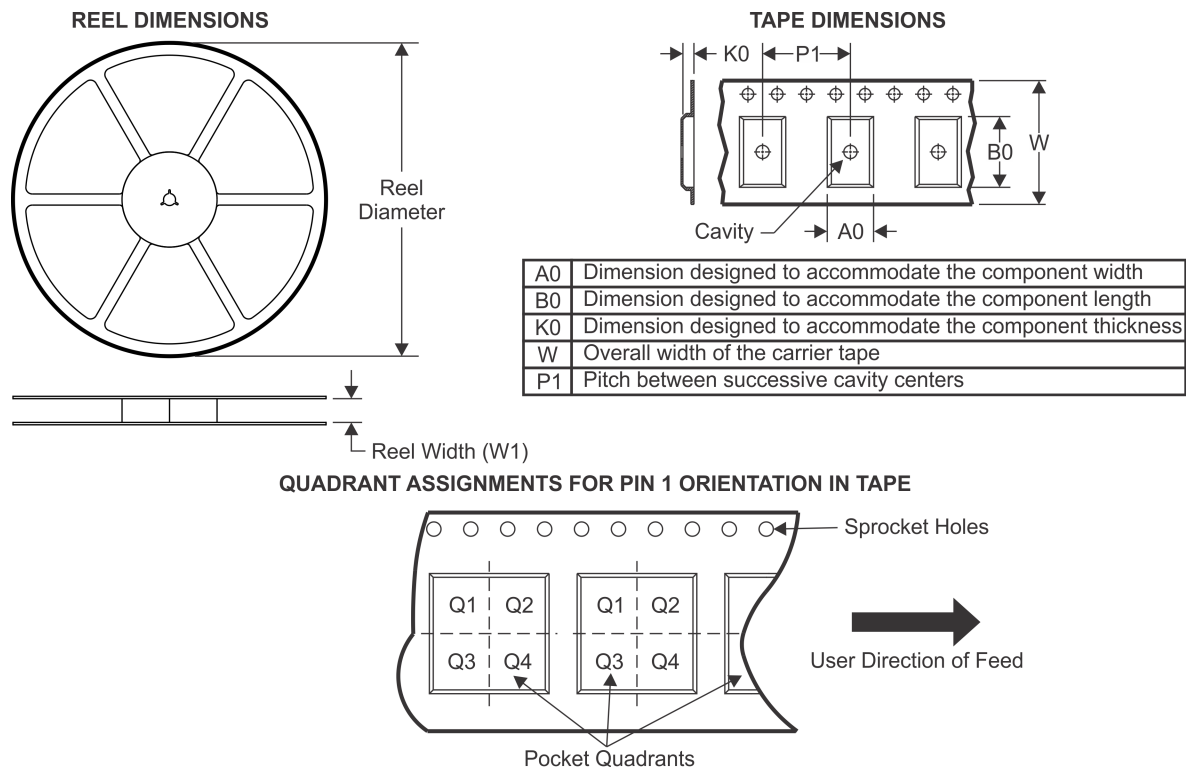
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

**OTHER QUALIFIED VERSIONS OF TCA9548A :**

- Automotive: [TCA9548A-Q1](#)

NOTE: Qualified Version Definitions:

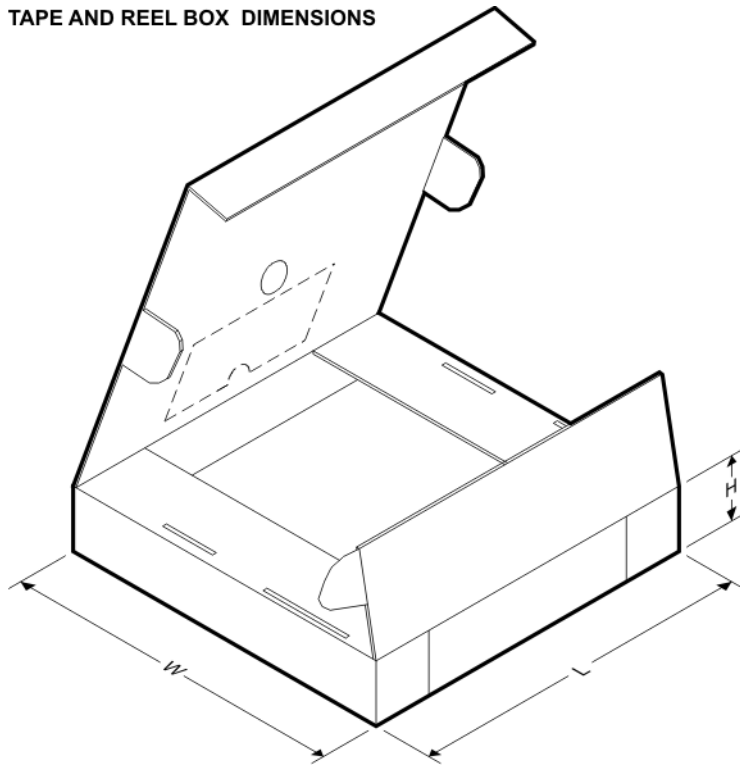
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

**TAPE AND REEL INFORMATION**


\*All dimensions are nominal

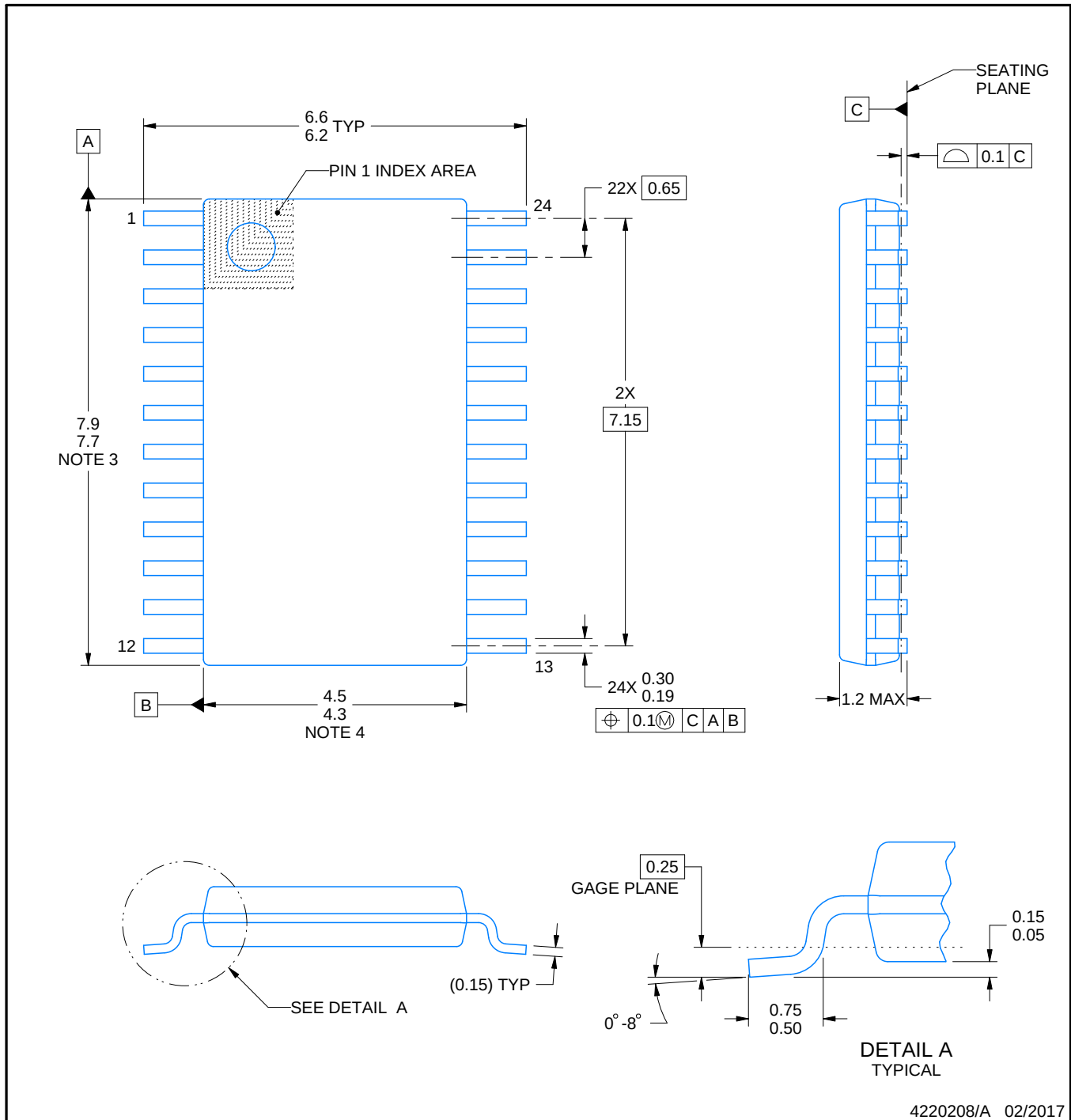
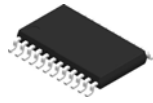
| Device        | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TCA9548AMRGER | VQFN         | RGE             | 24   | 3000 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q1            |
| TCA9548APWR   | TSSOP        | PW              | 24   | 2000 | 330.0              | 16.4               | 6.95    | 8.3     | 1.6     | 8.0     | 16.0   | Q1            |
| TCA9548ARGER  | VQFN         | RGE             | 24   | 3000 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q2            |

## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|---------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TCA9548AMRGER | VQFN         | RGE             | 24   | 3000 | 367.0       | 367.0      | 35.0        |
| TCA9548APWR   | TSSOP        | PW              | 24   | 2000 | 367.0       | 367.0      | 38.0        |
| TCA9548ARGER  | VQFN         | RGE             | 24   | 3000 | 367.0       | 367.0      | 35.0        |



4220208/A 02/2017

## NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

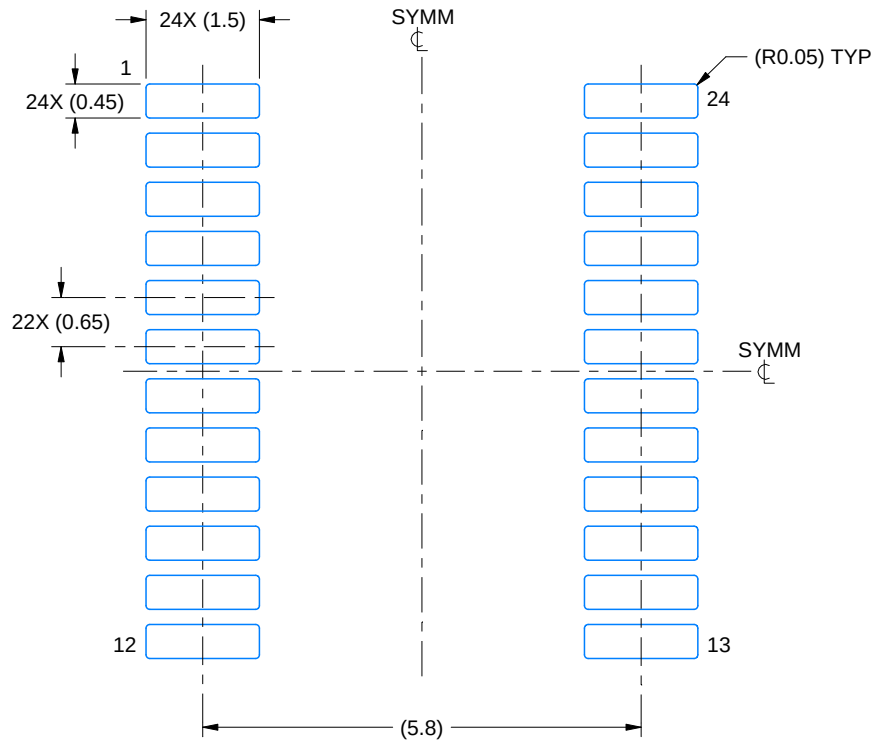


# EXAMPLE BOARD LAYOUT

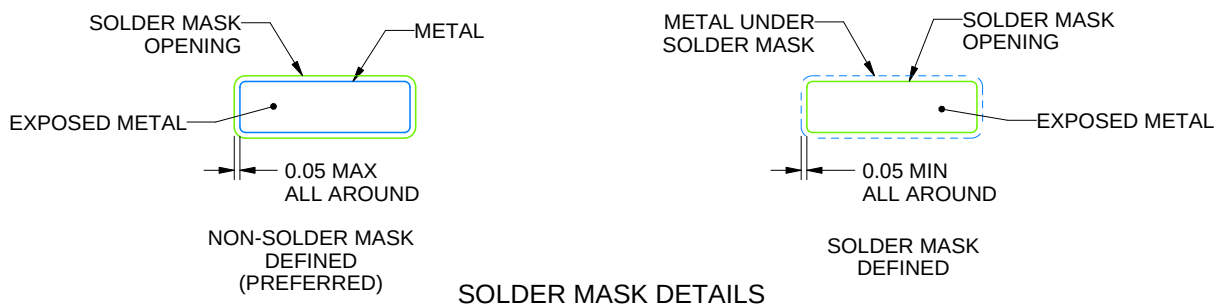
PW0024A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220208/A 02/2017

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

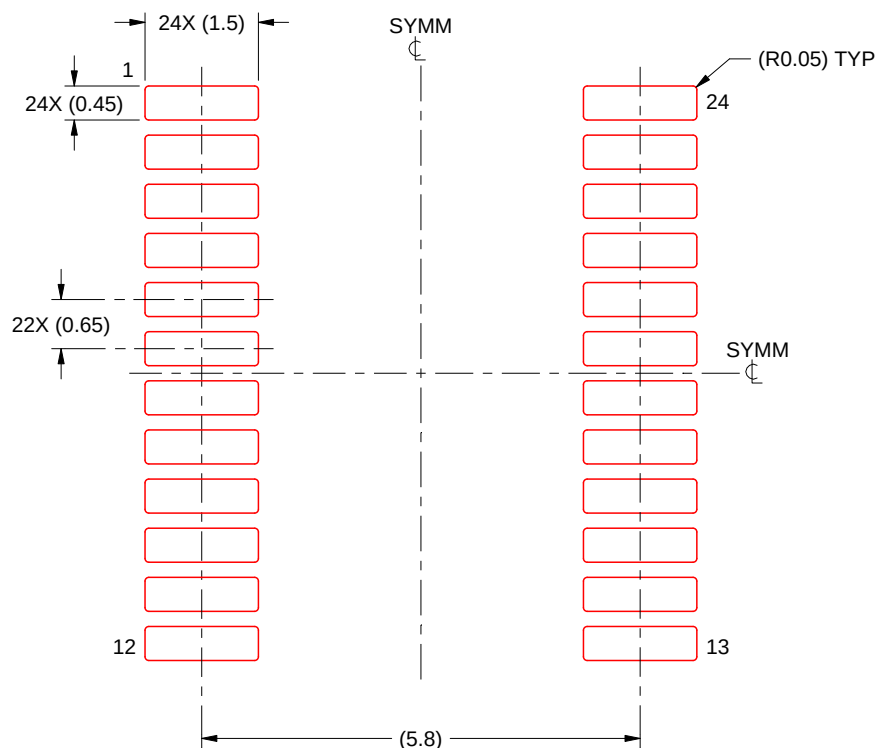
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

PW0024A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220208/A 02/2017

NOTES: (continued)

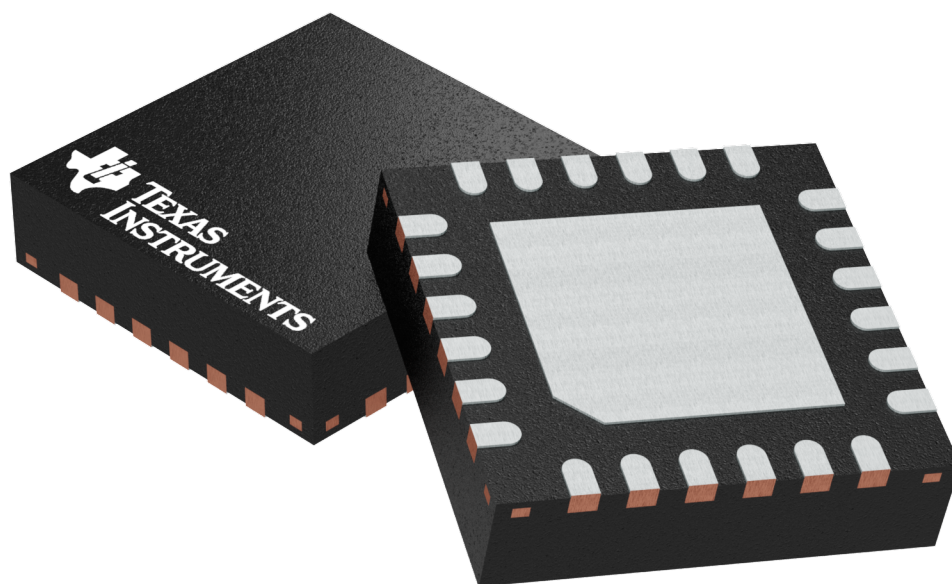
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

**RGE 24**

**GENERIC PACKAGE VIEW**

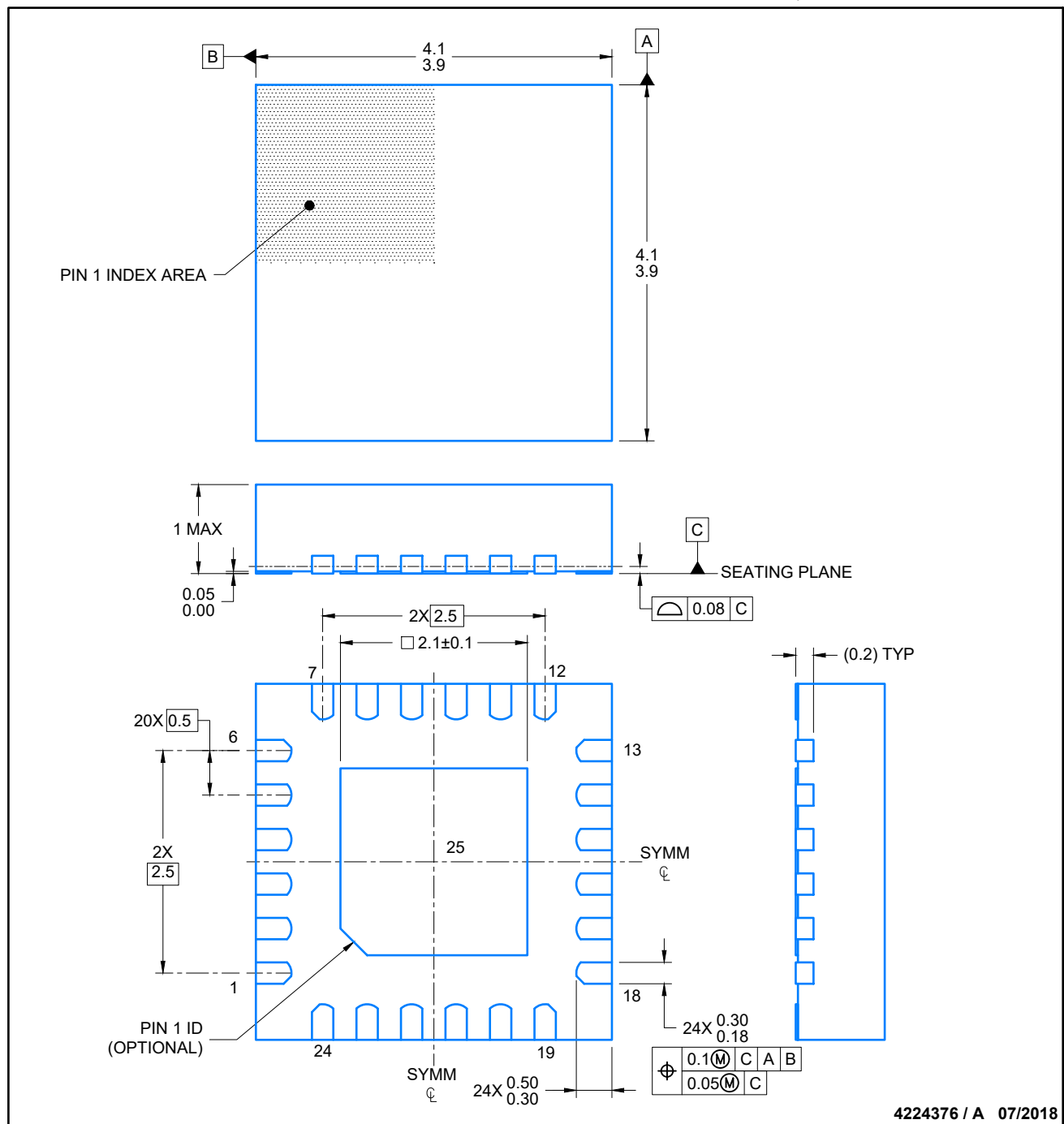
**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD



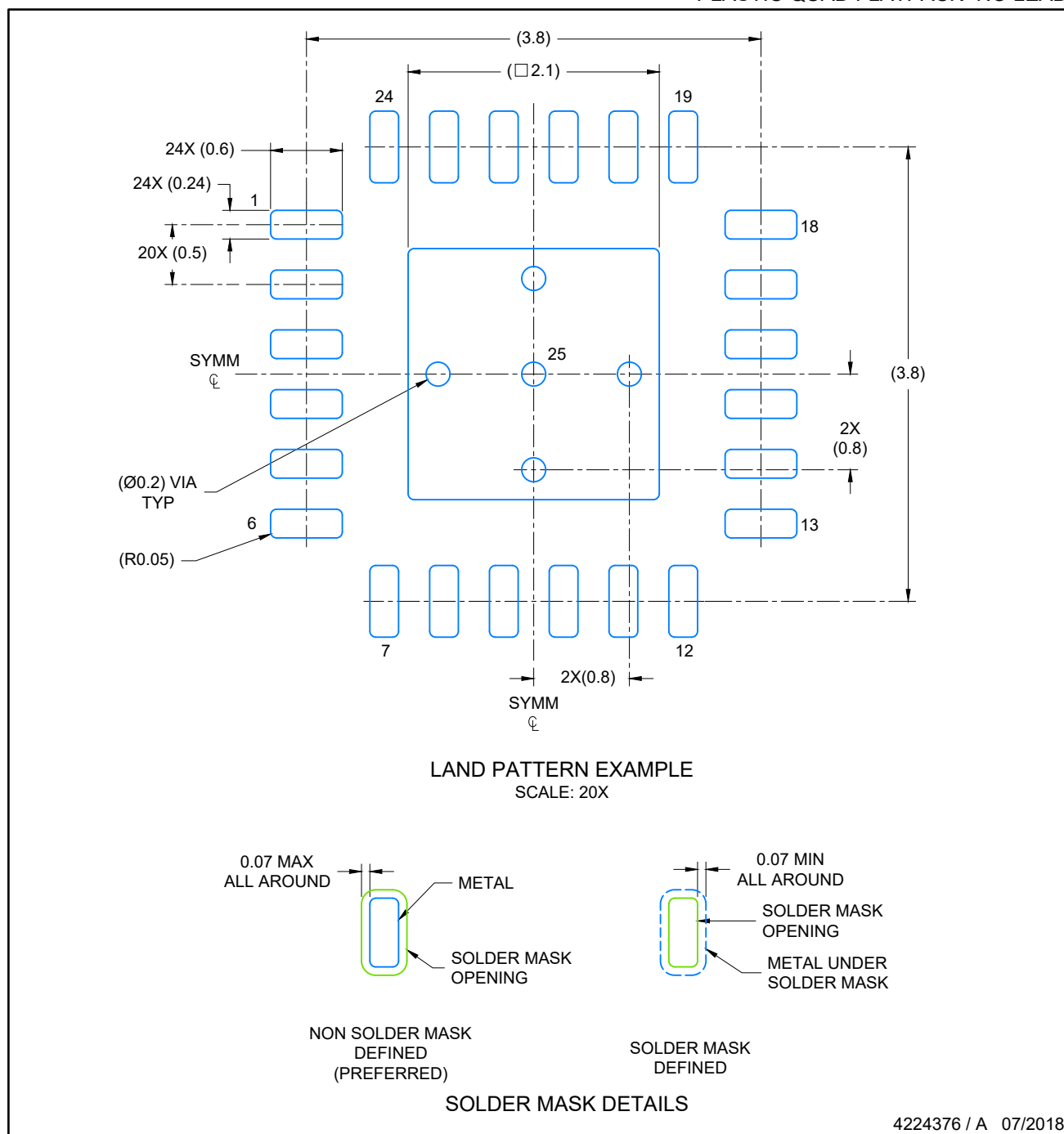
Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

4204104/H



## NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.



4224376 / A 07/2018

NOTES: (continued)

- This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/sluea271](http://www.ti.com/lit/sluea271)).
- Solder mask tolerances between and around signal pads can vary based on board fabrication site.

PLASTIC QUAD FLATPACK- NO LEAD

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